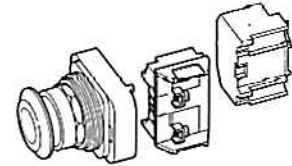


2-Position Push-Pull and Push-Pull/Twist Release and 3-Position Push-Pull Units, Illuminated

800T — FX T PH 16 R A1
a b c d e f g



a

Finger-Safe Guards	
Code	Description
Blank	No Guards
C	Guards on Terminals

b

Head Type	
Code	Description
FX	Mushroom Head
FXJ	Jumbo Mushroom Head

C

Operator Function			
2-Position 0			
Code	Description		
Blank	Push-Pull		
T	Push-Pull/Twist		
3-Position			
Operator Position			
Code			
M	★	‡	‡
N	★	†	★

★ Momentary‡ Maintained

d

Illumination Options	
Transformer	
Code	Description
P	Incandescent
PF	Flashing Incandescent
PH	LED

d (cont'd)

Illumination Options	
Full Voltage	
Code	Description
Q	Incandescent
QF	Flashing Incandescent
QH	LED
Resistor	
Code	Description
Q	Incandescent
Neon	
Code	Description
R	Neon
Dual Input	
Code	Description
D	Diode Type
DH	Diode Type LED
DT	Transformer — Relay Type
DTH	Transformer — Relay Type LED

e

Voltage	
Transformer	
Code	Description
36	48V AC 50/60 Hz
16	120V AC 50/60 Hz
26	240V AC 50/60 Hz
76	277V AC 50/60 Hz
46	480V AC 50/60 Hz
56	600V AC 50/60 Hz

e (cont'd)

Voltage	
Full Voltage	
Code	Description
06	6V AC/DC
12	12V AC/DC
24	24V AC/DC
32	32V AC/DC
48	48V AC/DC
10	120V AC/DC
13	130V AC/DC
20	240V AC/DC
Resistor	
Code	Description
11	120V AC/DC Resistor
Neon	
Code	Description
10	120V AC/DC
20	240V AC/DC
Dual Input	
Code	Description
16	120V AC
24	24V AC/DC

f

Color Cap	
Code	Color
X	No Cap
A	Amber
B	Blue
C	Clear
G	Green
R	Red
W	White

g

Contact Blocks	
Code	Description
Blank	No Contacts on operator
Standard	
D1	1 N.O.
D2	1 N.C.
D4	1 N.C.L.B.
A	1 N.O. - 1 N.C.
A1	1 N.O. - 1 N.C.L.B.
A2	2 N.O.
A5	2 N.C.L.B.
A7	1 N.C. - 1 N.C.L.B.
PenTUFF (Low Voltage)	
D1V	1 N.O.
D2V	1 N.C.
D4V	1 N.C.L.B.
AV	1 N.O. - 1 N.C.
Class 1, Div. 2/Zone 2	
Logic Feed	
D1R	1 N.O.
D2R	1 N.C.
AR	1 N.O. - 1 N.C.
A2R	2 N.O.
A4R	2 N.C.
Sealed Switch	
D1P	1 N.O.
D2P	1 N.C.
AP	1 N.O. - 1 N.C.
Stackable Sealed Switch	
D1Y	1 N.O.
D2Y	1 N.C.
A2Y	2 N.O.
A4Y	2 N.C.
AY	1 N.O. - 1 N.C.

Note: See Table 1 for Target description.

Table 1. Target Selection

2-Position			Contact Description	3-Position			g Code
g Code	Out	In		Out	Ctr.	In	
D1	O	X	N.O.	—	—	—	—
D4	X	O	N.C.L.B.	—	—	—	—
A/AV/AR/AP/AY	O	X	N.O.	O	O	X	A/AV/AR/AP/AY
	X	O	N.C.	X	O	O	
A1	O	X	N.O.	O	O	X	A1
	X	O	N.C.L.B.	X	X	O	
A2	O	X	N.O.	O	O	X	A2
	O	X	N.O.	O	O	X	
A5	X	O	N.C.L.B.	X	X	O	A5
	X	O	N.C.L.B.	X	X	O	
A4	X	O	N.C.	—	—	—	—
	X	O	N.C.	—	—	—	
—	—	—	N.C.	X	O	O	A7
	—	—	N.C.L.B.	X	X	O	

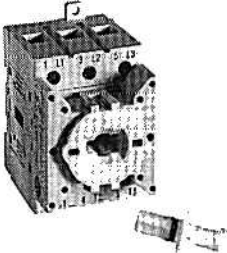
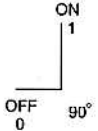
Note: X = Closed/O = Open

- 2-position push-pull and push-pull/twist-to-release devices with N.C.L.B. contacts meet EN-418 and IEC 60947-5-5 standards for emergency stop applications.
- Flashing lamps are only available for 6V full voltage units and all transformer units.
- LEDs available in red, green, amber, blue, and white. White LEDs only available in 6V and 24V full voltage and all transformer units. LED color must match lens color, except clear lens supplied with white LED and white lens supplied with amber LED. All LEDs except 120V have an internal shunt resistor for use with solid-state outputs.
- Neon is available in amber or clear.

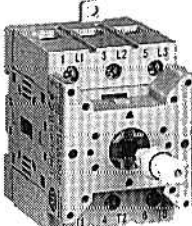
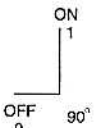
- Diode type dual input provides circuit isolation via opposing diodes. Not recommended for use with solid-state outputs and neon indicators.
- 32V and 130V are LED only. 240V is incandescent only.
- 120V LED is AC only and does not contain internal shunt resistor. For AC/DC and internal shunt resistor, order 130V AC/DC LED (code 13).
- Dual input diode only.
- Not valid with head Type J.

Allen-Bradley

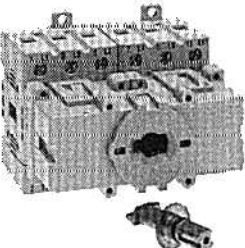
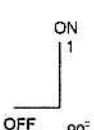
OFF-ON Base-Mounted 3-Pole Switch (includes operating shaft) (Actuators listed on pages 4-4 and 4-5)

Switch Body (Includes Shaft)	Function Switching Angle	Contact Target Configuration X = Contact Closed O = Contact Open			Maximum Load Rating (A)	AC23A Rated Power [kW] at 690V AC 50 Hz	HP @ 480V AC 60 Hz 3 Ø	Cat. No.
		No. of Circuits	Handle Position					
			OFF/0 	ON/1 				
		1	O	X	25	11	10	194E-A25-1753
		2	O	X	32	15	15	194E-A32-1753
		3	O	X	40	18.5	20	194E-A40-1753
					63	22	25	194E-A63-1753
					80	37	50	194E-A80-1753
					100	45	50	194E-A100-1753

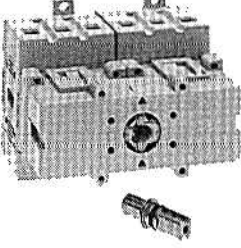
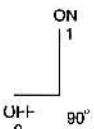
OFF-ON Front-Mounted 3-Pole Switch (includes operating shaft) (Actuators listed on pages 4-4 and 4-5)

		1 2 3	O O O	X X X	25	11	10	194E-E25-1753
					32	15	15	194E-E32-1753
					40	18.5	20	194E-E40-1753
					63	22	25	194E-E63-1753
					80	37	50	194E-E80-1753
					100	45	50	194E-E100-1753

OFF-ON Base-Mounted 6-Pole Switch (includes operating shaft) (Actuators listed on pages 4-4 and 4-5)

		1 2 3 4 5 6	O O O O O O	X X X X X X	25	11	10	194E-A25-1756
					32	15	15	194E-A32-1756
					40	18.5	20	194E-A40-1756
					63	22	25	194E-A63-1756
					80	37	50	194E-A80-1756
					100	45	50	194E-A100-1756

OFF-ON Front-Mounted 6-Pole Switch (includes operating shaft) (Actuators listed on pages 4-4 and 4-5)

		1 2 3 4 5 6	O O O O O O	X X X X X X	25	11	10	194E-E25-1756
					32	15	15	194E-E32-1756
					40	18.5	20	194E-E40-1756
					63	22	25	194E-E63-1756
					80	37	50	194E-E80-1756
					100	45	50	194E-E100-1756



User & Technical Manual
RZ-05A, RZ-06A & RZ-08A Series
Rev 3.0



CAUTION

**Read Manual Before Installation.
For Questions, Please Call ISC at
(800)-836-7472**

To Be Professionally Installed Only



Title: RZ-05A, RZ-06A & RZ-08A Air Conditioner Technical Manual

Department: Engineering

Objective: To provide important information for maintenance, diagnostics and advance operations of the DELTA-T RZ-05A, RZ-06A & RZ-08A Series Air Conditioners.

Revision History:

Rev	Date	Owner	Description of Changes
1.6	01-16-13	B. Slotnick	New Format Initial Release
1.7	02-05-13	B. Slotnick	Corrected heater options
1.8	02-26-13	B. Slotnick	Added power terminal block
1.9	08-09-13	B. Slotnick	Added Appendix A
2.0	01-03-14	B. Slotnick	Update Appendix A photo
2.1	01-15-14	B. Slotnick	Update Adjustable Temperature Probe length
2.2	08-04-14	B. Slotnick	Delete Celsius programming flash drive key
2.3	01-23-15	B. Slotnick	Update product drawings and system schematic
2.4	03-24-15	B. Slotnick	Parts list: Added high pressure switch; updated filter information
2.5	07-25-16	B. Slotnick	Removed 480 V SCCR rating. Added circuit breaker recommendation.
2.6	01-17-17	B. Slotnick	Corrected unit weights
2.7	07-03-17	B. Slotnick	Changed Outdoor Option to Remote Controller
2.8	08-31-17	B. Slotnick	Updated replacement temperature probe information
2.9	10-05-17	B. Slotnick	Added replacement parts to warranty statement & new label template
3.0	01/26/18	B. Slotnick	Updated unit specifications

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TABLE OF CONTENTS

Overview	4
Inspecting the Equipment	4
Unpacking the Air Conditioner	4
Moving the Air Conditioner	5
Unit Label	5
Operation	6
Preliminary Testing	6
Mounting the Air Conditioner	7
Unit Specifications	8
Options	10
System Faults	13
Digital Temperature Controller Programming	13
Preventative Maintenance	16
Field Serviceable Parts	17
Safety Information	17
Troubleshooting Guide	18
Physical Dimensions	21
Electrical Schematic	23
Warranty Information	24
Return Material Authorization (RMA) Procedure	25
Appendix A – Power Terminal Block	26



Overview

Thank you for your purchase of the DELTA-T Special Purpose Air Conditioner. Our air conditioning equipment is carefully designed to cool and dehumidify the air in electronic component enclosures. ISC has designed air conditioners for all types of electronic equipment enclosures providing capacity from 1,000 to 20,000 BTUH.

This manual will guide you through the installation, maintenance, diagnostics and advance operations of the RZ-05A, RZ-06A & RZ-08A Series Air Conditioners. This manual contains important information for the end-user who installs, maintains and/or operates the RZ-05A, RZ-06A & RZ-08A Air Conditioner.

Technical Support

By Phone: 972-964-2700 / 800-836-7472

(Monday – Friday, 7:30 am – 5:00 pm Central Time)

By Email: iscinfo@iscsales.com

Our goal is to have continuous improvement for both our equipment and our documentation. We rely and appreciate your feedback to help us achieve our goal. Our technical support team is glad to work with you if you require additional technical information not provided in this manual.

Inspecting the Equipment

DELTA-T air conditioning equipment is designed, manufactured and packed to prevent damage from normal handling, shock and vibration during shipment. It is necessary to inspect your equipment upon receipt to ensure that there is no visual or hidden damage.

Upon receipt of the equipment, ensure that it is always top side up as indicated by “THIS SIDE UP” labels. This ensures that the compressor oil is not displaced.

Note: If the equipment has been on its side it must be put upright for **at least 24 hours** before installing or running to ensure the compressor oil has returned to the compressor.

All physical damage to packing or signs of damage to the equipment must be noted on the freight bill of lading. Packages must be opened after receipt and inspected for any internal or concealed damage to the equipment and to verify proper count and order fulfillment. Delivery of without the pallet, not top side up, other freight on top, damaged or wet should be refused.

Unpacking the Air Conditioner

Always keep air conditioner top up as shown by the packaging label and arrows. If the unit is to be transported after initial unpacking, place air conditioner back in original packing to prevent damage.

For shipment by UPS or freight carrier, repack as received and re-band to the pallet.

Note: Shipping without being strapped to the pallet may result in tipping and damage, this will void the warranty.



Moving the Air Conditioners

Read this section completely before running or installing your DELTA-T air conditioning equipment.

Note: You will need to perform a Preliminary Test before mounting the air conditioner. Refer to the Preliminary Test section in this manual for instructions on how to run this test. You will also need to prepare the enclosure that is to be cooled for mounting in accordance with this manual and the template supplied.

Note: DELTA-T air conditioning equipment must be **mounted vertically** with a **minimum of 3"** air space for condenser air return and supply. If necessary, equipment may be mounted at up to a 5 degree angle (rotary compressors) without damaging the compressor.

If air conditioner is to be shipped or transported at any time; pack in original packaging and strap to pallet to prevent damage. Air conditioner must be kept upright at all times. Air conditioners are not designed to be shipped attached to an equipment enclosure. Air conditioners shipped which have internal damage due to shipping while attached to enclosures are warranty voided.

Unit Label

Each air conditioner has a unit label, be sure to record the data from the label to the template below and keep this information in a safe place for warranty and ordering parts. To prevent damage to equipment, electrical panel and wiring, and to prevent personal injury, assure that the power source is compatible with the equipment before operating. When recharging, refrigeration type and amount must be the same as shown on the unit label or air conditioner will not operate properly or may be damaged and may result in the warranty voided and UL listing voided.

Voltage: _____	Frequency: _____ Hz	Phase: _____
Current Rating (A): _____	Cooling: _____	Heat: _____
Max Fuse: _____ Min.	Circuit Ampacity: _____	
Refrigerant: _____	Ounces: _____	
Design Pressure: _____	High Side PSI: _____	Low Side PSI: _____
NEMA Type: _____	IP: _____	
Model: _____		
S/N: _____		

Operation

DELTA-T air conditioners will lower (or increase as necessary) the temperature inside an enclosure to ensure its proper operational temperature. Our air conditioners, when sized properly, will provide cooling or heating automatically controlled by the temperature digital controller.

DELTA-T air conditioners operate as a "closed loop" system with no exposure or introduction of outside air. This ensures that the enclosure is separated from and is not contaminated with ambient air, dirt, chemicals, dust, moisture or foreign matter so that sensitive enclosure components are protected and are kept at your required operational temperature.

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An air conditioner is designed to dehumidify and extract heat from an area, or provide heat to an area. The cooling is done using a simple refrigeration cycle. A product of this simple refrigeration cycle is excess humidity that condenses to a liquid.

DELTA-T air conditioners are designed using an advanced refrigeration cycle and are equipped with a Condensate Removal System that changes the excess humidity liquid into a vapor which is then vented to the atmosphere. In the event of excess water vapor where the enclosure door has been left open there is an overflow hole on the bottom of every unit. Please contact ISC Inc. if you encounter excess water coming from your air conditioner.

Preliminary Testing

Before mounting the air conditioner to the enclosure, test for proper operation. Follow the steps below prior to installation.



WARNINGS

The air conditioner must sit upright for 24 hours to assure the compressor oil has drained down to the compressor. Check the unit label to assure the electric power available to the air conditioner is the proper voltage and phase. Check the electric power source for proper ground wire and neutral wire installation per 2008 NEC. Assure that the electric power is protected by a circuit protection device; refer to the Unit Specification section in this manual for proper circuit protection sizing.

- Connect a power source to the air conditioner to start the evaporator fan and, if desired, change the set point. Refer to Digital Temperature Controller Programming section on this manual for details on changing the set point. Factory default settings should be satisfactory; however you may want to adjust heat, cooling and alarm set points for your specific application.
- The condenser fan is controlled by the refrigerant high heat pressure and will turn on later when the operating pressure builds up. This feature allows low ambient operation and reduces current inrush at initial power on.
- Run the air conditioner for 15 minutes; during this time the condenser fan will turn on after the condenser coil warms up. **Note: Ambient Temperature must be at least 75°F.**
- Turn the unit off if the equipment makes any unexpected or hard mechanical noises or vibrations and refer to the troubleshooting guide in this manual.
- The supply cold air should be at least 10°F colder than the return air entering the unit. If it is not, refer to the troubleshooting guide in this manual;
- When you are satisfied that the unit is operating properly, turn unit off, disconnect the power and mount the unit on the enclosure in accordance with the Mounting the Air Conditioner section in this manual.

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Mounting the Air Conditioner

Using the template supplied, determine where the air conditioner is to be mounted and assure that all required cuts and holes will not interfere with or damage the enclosure or its contents. Assure that there is a 3" clearance between walls / obstructions and the air conditioner for the condenser supply air and return air flow so that it is not restricted. Restricted condenser air flow will affect the air conditioner's performance. Mount the air conditioner high on the enclosure in order to cool the hot air in the top of the enclosure. Position the unit where the cold air can circulate across the width of the enclosure to cool it all the way across.

Once proper mounting placement is determined, turn the enclosure equipment off, if possible, to prevent damage. Drill and cut the holes as indicated on the mounting template. Install insulation gasket as required to ensure an air tight closed loop seal. Be cautious not to let any cutting debris fall into the enclosure.

Hang the air conditioner on the Easy Hang Tabs and from inside the enclosure use the fasteners supplied to attach the air conditioner to the enclosure. Ensure that these fasteners are tight in order to prevent the unit from falling off the enclosure. These fasteners should be checked periodically to ensure that they have not become loose due to vibration.

The air conditioner is provided with a power cord or a power terminal block. Ensure that the power supplied is compatible with the air conditioner's power requirements. Properly connect the unit to a power circuit that meets the equipment requirements and provide a circuit protection device based on the Unit Specifications section of this manual. See Appendix A for Wago power terminal block instructions.

After mounting the air conditioner, replace/close the enclosure door and start air conditioner; test for air leaks to assure a proper closed air loop seal and run test the unit to assure proper operation after mounting. If any cold air leaks are found, check for proper mounting and apply silicone-free Lexel seal if leaks persist. Assure that the air conditioner's condenser inlet air filter is installed properly and cleaned regularly.



CAUTION

If mounting the air conditioner to the enclosure door, confirm with the enclosure manufacturer that the door's hinges will support the air conditioner's added weight (see equipment specifications). Ensure that when the door is fully open that the enclosure will not topple over due to the off-center load.



Unit Specifications

The following table provides electrical and refrigeration specifications for the RZ-05A, RZ-06A & RZ-08A Series Air Conditioners.

RZ-05A Series

Nominal Capacity Rating: 5,000 BTUH

Standard Maximum Ambient: 125°F

Unit Weight: 120VAC – 97lbs, 230VAC – 92lbs, 480VAC – 136lbs

Model	Operating Voltage Range (Volts)	Inrush Current (Start Up Current) (Amps)	Loading Current (Running Current) (Amps)	SCCR (Short Circuit Current Rating) (Amps)	Recommended Fuse Selection (Amps)	KVA Rating	Refrigerant Type	Refrigerant Amount (ounces)	Low/High Unit Pressures (psi)
RZ-05A -126	110-120	23.42	7.26	*2	12 Amp Class CC, CCMR Time Delay *	0.85	R422d	12.50	98 / 314
RZ-05A -236	220-240	19.15	3.76	*2	10 Amp Class CC, CCMR Time Delay *	0.7	R422d	12.50	76 / 322
RZ-05A -486	440-480	9.18	1.5	*2	5 Amp Class CC, CCMR Time Delay *	0.7	R422d	12.50	76 / 322

* HACR Type Circuit Breakers or Time-delay Fuses with the following electrical characteristics are recommended. Fast-acting fuses are not recommended.

% of Ampere Rating	Opening Time
110 %	15min Minimum
125 %	1hr Maximum
200 %	12sec Minimum, 4min Maximum

*2 SCCR rating is based on the SCCR rating for the circuit protection device installed in the panel / enclosure per UL50 & UL508a to protect the AC unit. Typically 100KA – 200KA for Time-Delay Fuses.



RZ-06A Series

Nominal Capacity Rating: 6,000 BTUH

Standard Maximum Ambient: 125°F

Unit Weight: 120VAC – 97lbs, 230VAC – 98lbs, 480VAC – 142lbs

Model	Operating Voltage Range (Volts)	Inrush Current (Start Up Current) (Amps)	Loading Current (Running Current) (Amps)	SCCR (Short Circuit Current Rating) (Amps)	Recommended Fuse Selection (Amps)	KVA Rating	Refrigerant Type	Refrigerant Amount (ounces)	Low/High Unit Pressures (psi)
RZ-06A -126	110-120	42.41	7.83	*2	25 Amp Class CC, CCMR Time Delay *	1.0	R422d	18.00	97 / 377
RZ-06A -236	220-240	21.15	4.80	*2	12 Amp Class CC, CCMR Time Delay *	0.9	R422d	18.00	103 / 343
RZ-06A -486	440-480	10.13	1.70	*2	5 Amp Class CC, CCMR Time Delay *	0.9	R422d	18.00	103 / 343

RZ-08A Series

Nominal Capacity Rating: 8,000 BTUH

Standard Maximum Ambient: 125°F

Unit Weight: 120VAC – 102lbs, 230VAC – 103lbs, 480VAC – 142lbs

Model	Operating Voltage Range (Volts)	Inrush Current (Start Up Current) (Amps)	Loading Current (Running Current) (Amps)	SCCR (Short Circuit Current Rating) (Amps)	Recommended Fuse Selection (Amps)	KVA Rating	Refrigerant Type	Refrigerant Amount (ounces)	Low/High Unit Pressures (psi)
RZ-08A -126	110-120	42.41	7.83	*2	25 Amp Class CC, CCMR Time Delay *	1.0	R422d	19.00	188 / 345
RZ-08A -236	220-240	21.15	4.80	*2	12 Amp Class CC, CCMR Time Delay *	0.9	R422d	19.00	188 / 410
RZ-08A -486	440-480	10.13	1.80	*2	5 Amp Class CC, CCMR Time Delay *	0.9	R422d	19.00	188 / 410

* HACR Type Circuit Breakers or Time-delay Fuses with the following electrical characteristics are recommended. Fast-acting fuses are not recommended.

% of Ampere Rating	Opening Time
110 %	15min Minimum
125 %	1hr Maximum
200 %	12sec Minimum, 4min Maximum

*2 SCCR rating is based on the SCCR rating for the circuit protection device installed in the panel / enclosure per UL50 & UL508a to protect the AC unit. Typically 100KA – 200KA for Time-Delay Fuses.

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Options

The RZ-05A, RZ-06A & RZ-08A Series Air Conditioners may be ordered with the following options. Review the list below for the specifications and functions of the option(s) that apply to your unit.

Built-in Heater

This unit is configurable to include a Built-in Heater to increase the temperature of your enclosure in lower temperature conditions. The digital controller is programmed with an individual heater set point to maintain your enclosure at your desired operating temperature.

Heater capacities are available in 500 Watt and 1000 Watt.

Default Settings:

- 55°F Heat Set Point
- 3°F Differential

At the default settings the heater turns on at 52°F and operates until temperature has reached 55°F. Refer to Digital Temperature Controller Programming section for details on changing these parameters.

Low Ambient

This option consists of circuitry to protect the compressor in a low ambient temperature environment. The low ambient package is designed to provide an optimum constant low current in the compressor motor during the non cooling state. This option provides internal heat to the compressor to protect it from stalls and hard starts during long periods of compressor off state.

Dry Contact

This option provides an early warning capability to your enclosure to aid in the protection of your expensive equipment. The Dry Contact is programmed with an individual dry contact set point to monitor the enclosure temperature limit that you require. When the enclosure temperature has exceeded the maximum limit, the Form C gold plated contact will activate to notify you that there is a problem in achieving your cooling set point.

Default Settings:

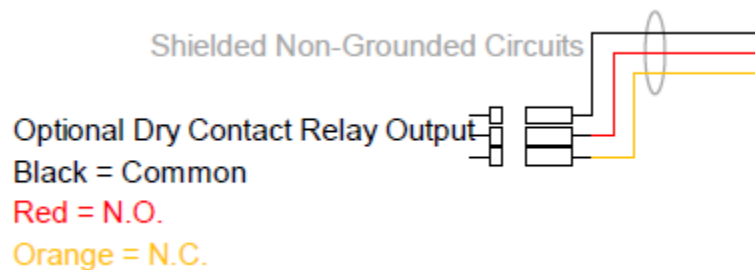
- 105°F Dry Contact High Temperature Alarm Set Point
- 2°F Recovery Differential
- 3min Temperature Alarm Delay

At the default settings the Dry Contact initial conditions are normally open and/or normally closed. When the unit is powered on and the temperature exceeds 105°F for a period longer than 3 minutes the Dry Contact will activate. At this time the normally open contact will close and the normally closed contact will open. This state will remain until the temperature has decreased to 103°F.

Note: The dry contact is not a power status indicator, rather a high temperature alarm alerting you that the unit is unable to meet the cooling set point.

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Refer to Advanced Digital Temperature Controller Programming section on page 8 for details on changing these parameters.

Celsius Programming

Celsius Programming option allows the user to control the DELTA-T Air Conditioner in SI units. This feature can be programmed into the digital temperature controller at time of manufacturing.

External Heat Output

This option allows you to add an external heater to be located anywhere in your enclosure. These external heaters are ideal for focusing on individual sections of the enclosure without having a large capacity Built-in Heater.

External Heater capacities available are 100 Watt, 150 Watt, 200 Watt, 250 Watt, 300 Watt, 400 Watt, 550 Watt, 650 Watt & 950 Watt

Default Settings:

- 55°F Heat Set Point
- 3°F Differential

At the default settings the heater output turns on at 52°F and operates until the temperature has reached 55°F. Refer to Advanced Digital Temperature Controller Programming section on this manual for details on changing these parameters.

Special Programming

All DELTA-T Air Conditioners have a Special Programming option for selecting unique pre in programmed settings for special requirements. Although most parameters are accessible in our digital controller, this option allows you to receive a unit preset to your requirements without the need to manually configure the unit. Refer to Advanced Digital Temperature Controller Programming section for details on all default settings.

High Ambient

Most of our air conditioners have capabilities for high ambient application upgrades from our Standard Maximum Ambient temperature of 125°F. With this option we can optimize the Air Conditioner during manufacturing to achieve improved performance at high ambient conditions.

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Open Door Kill Switch

This option provides you with an adjustable and mountable safety switch for your electronics enclosure. This switch will disable power to the air conditioner when your electronics enclosure door is open allowing you to work on your enclosure without the risk of unintentionally powering the unit.

Adjustable Temperature Probe

The adjustable temperature probe provides you with a 12' temperature probe that can be installed anywhere in your enclosure. This allows you to have the flexibility to monitor and maintain the temperature at any specific location of your enclosure.

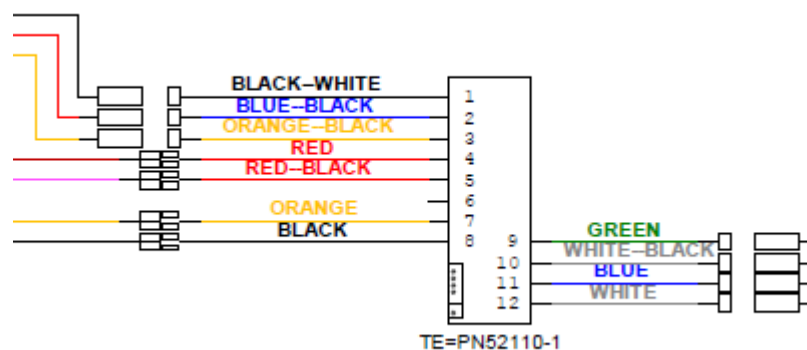
Ethernet/IP Controller Output

This option provides the ability to communicate with the DELTA-T Air Conditioner controller via the Ethernet/IP protocol. You can monitor and control all settings from a remote location with a PLC.

Full specifications will be provided with this option. For full details refer to the Ethernet/IP Option Specifications included with the Air Conditioner.

Remote Controller

When having our Programmable Digital Controller on the face of our Air Conditioner does not fit your application, ISC offers the Remote Controller Option. This option moves the controller into your enclosure using a 10' cable along with a universal, 16 gauge powder coated steel mounting bracket. The Air Conditioner will have no openings on the face of the unit. Below is partial schematic showing the Remote Controller option wiring connections to the digital controller.





System Faults

If any of the critical control parameters exceed limits, the compressor is turned off and an alarm condition is indicated on the front panel.

There are three main conditions that can shut the AC compressor down:

- The condenser high temperature alarm
- The evaporator coil alarm
- The compressor thermal overload

In the event that the condenser coil overheats, a condenser high temperature alarm, **HA2**, will flash on the display of the digital temperature controller and the compressor will turn off after a 3 minute time delay.

In the case that the evaporator coil ices up or there is a leak, an evaporator coil alarm, **CA**, will flash on the display of the digital temperature controller and the compressor will turn off after a 2 minute time delay.

A thermal overload protects the compressor against faults. In the event that the snowflake status LED is not flashing on the display and the compressor is not running, an internal fault may have occurred on the unit. Refer to the Troubleshooting Guide in this manual.

Digital Temperature Controller Programming

For the end user this section will provide all the necessary information to view and change all functions and parameters available to you.



HOW TO SEE THE SETPOINT

1. Press and immediately release the **SET** key. The display will show the setpoint value.
2. Press and immediately release the **SET** key or wait for 15 seconds to display the enclosure temperature again.

HOW TO CHANGE THE SETPOINT

1. Press the **SET** key for more than 2 seconds to change the setpoint value.
2. The value of the setpoint will be displayed and the "F" LED starts blinking.
3. To change the **SET** value, press the **UP** or **DOWN** arrow key within 15 seconds.
4. To store the new setpoint value, press the **SET** key again or wait 15 seconds.

Note: The set value is stored even when the procedure is exited by waiting for the time-out to expire.

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HOW TO ACCESS HIDDEN MENU 1

1. Enter the Programming mode by pressing the **SET + DOWN** arrow keys simultaneously for 3 seconds. The "F" LED starts blinking.
2. Select the required parameter by pressing the **UP** or **DOWN** arrow key. Press the **SET** key to display its value.
3. Use the **UP** or **DOWN** arrow key to change its value.
4. Press **SET** to store the new value and move to the following parameter.

To exit: Press **SET + UP** arrow keys or wait 15 seconds without pressing a key.

Note: The set value is stored even when the procedure is exited by waiting for the time-out to expire.

HOW TO ACCESS HIDDEN MENU 2

1. Enter the Programming mode by pressing the **SET + DOWN** arrow keys simultaneously for 3 seconds. The "F" LED starts blinking.
2. Release the keys, then push the **SET + DOWN** arrows keys simultaneously, again for more than 7 seconds. The **Pr2** label will be displayed immediately followed by the **HY** parameter.
3. Select the required parameter by pressing the **UP** or **DOWN** arrow key. Press the **SET** key to display its value.
4. Use the **UP** or **DOWN** arrow key to change its value.
5. Press **SET** to store the new value and move to the following parameter.

To exit: Press **SET + UP** arrow keys or wait 15 seconds without pressing a key.

Note: The set value is stored even when the procedure is exited by waiting for the time-out to expire.

HOW TO LOCK THE KEYPAD

1. Press and hold the **UP** and **DOWN** arrow keys simultaneously for more than 3 seconds.
2. The **POF** message will be displayed and the keyboard will be locked. At this point, it will be possible only to see the setpoint or the MAX or MIN temperature stored.
3. If a key is pressed for more than 3 seconds the **POF** message will be displayed.

HOW TO UNLOCK THE KEYPAD

1. Press and hold the **UP + DOWN** arrow keys simultaneously for more than 3 seconds until the **Pon** message is displayed.

Functions and Parameters

Function	Operation	Parameter	Menu Location	Factory Default	Operating Range	Unit
Cooling Setpoint	Set Desired Cooling Temperature	SET	Main	90	70 - 95	°F
Cooling Differential	Control Hysteresis	Hy	Hidden Menu 1	5	1 - 45	°F
Compressor Protection	Anti short-cycle delay	AC	Hidden Menu 2	5	0 - 50	min
Probed Displayed	Select Display Probe P1 – Enclosure Temperature P2 – Condenser Temperature	Lod	Hidden Menu 2	P1	P1 / P2	N/A
Auxiliary Setpoint	Setpoint for Dry Contact *1 / Heater	SAA	Hidden Menu 2	105 / 55	-67 - 302	°F



Function	Operation	Parameter	Menu Location	Factory Default	Operating Range	Unit
Auxiliary Differential	Heater or Dry Contact Hysteresis *1	Shy	Hidden Menu 2	3	1 - 45	°F
Maximum Temperature Alarm	Alerts with "HA" when maximum temperature has been exceeded	ALU	Hidden Menu 2	105	45 - 302	°F
Minimum Temperature Alarm	Alerts with "LA" when minimum temperature has been exceeded	ALL	Hidden Menu 2	45	-67 - 105	°F

*1 **Note:** When both Heater and Dry Contact options are included, these parameters only control Heater Settings. Dry Contact is controlled by ALU.

Alarms Signals

Message	Cause	Outputs
"P1"	Enclosure Probe Failure	Start and stop of the compressor after 15 minute intervals *2
"P2"	Condenser Probe Failure	No Output *2
"HA"	Maximum Temperature Alarm: P1 > 105°F for 3 minutes (after initial 30 min. start up delay)	No Output *2
"LA"	Minimum Temperature Alarm: P1 < 45°F for 3 minutes (after initial 30 min. start up delay)	No Output *2
"HA2"	Condenser High Temperature Alarm: P2 > 145°F for 3 minutes	Compressor Off *2
"LA2"	Condenser Low Temperature Alarm: P2 < 50°F for 3 minutes	No Output *2
"CA"	Evaporator Coil Alarm: Low Pressure for 2 minutes	Compressor Off *2
"noP"	Invalid Probe Selected: Low Evaporator Coil	No Output, Display Only

*2 **Note:** When both Heater and Dry Contact options are included, Dry Contact (Alarm Contact) Relay is activated.

ALARM RECOVERY

If the compressor is turned off due to an alarm, it will be able to restart after the **AC** anti short-cycle time delay. (Parameter AC = 5minutes)

Probe alarm **P1** is cleared with no time delay after the fault is cleared.

Probe alarm **P2** starts and stops with no time delay.

Temperature alarm **HA** automatically clears as soon as the temperature returns to below 105°F (the **ALU** setpoint) -2°F = 103°F.

Temperature alarm **LA** automatically clears as soon as the temperature returns to above 55°F (the **ALL** setpoint) +2°F = 47°F.

Temperature alarm **HA2** automatically clears as soon as the condenser coils temperature returns to below 135°F.

Temperature alarm **LA2** automatically clears as soon as the condenser coils temperature returns to above 60°F.

Temperature alarm **CA** automatically clears as soon as the pressure returns to normal.

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Preventative Maintenance

Air conditioners require regular cleaning of the condenser air inlet filter. Wash filter often if washable using proper cleaning compound and back flushing the dirt out, or replace if not washable whenever it appears physically dirty. Restriction to the flow of air over the condenser coil will degrade the performance of the equipment, cause it to overheat, reduce cooling and can damage the compressor. The overload switch in the compressor may cause the compressor to cycle if the condensing coil pressure becomes too high due to air restriction. A further safety feature is the refrigerant high pressure cut off switch which will stop the compressor if the condensing pressure becomes too high due restricted condenser air flow or some other cause.

Restricted air flow due to neglecting a dirty filter will cause unit to repeatedly turn off and then turn back on after it cools off. An operation in this safety cycling mode will eventually damage the equipment and void the warranty.

Refer to Field Serviceable Parts section in this manual for details on parts that can be changed to help increase the uninterrupted life of the Air Conditioner.

Air Filter Service

Remove the air conditioner filter and wash if washable or replace if not washable. Dry thoroughly before replacing. DELTA-T Air Filter Treatment Spray in aerosol cans or spray bottle is recommended to enhance air filter efficiency. Replacement filters are readily available from ISC Customer Service. DO NOT operate the air conditioner without the air filter. Such operation will allow the condenser coil to become dirty and loose efficiency, causing overheating and diminished cooling capacity. Dirty condensing coils must be back-flush cleaned using proper commercial coil cleaning compounds and thorough back-flush rinsing. Refer to directions on the cleaning compounds selected. Acid wash is not recommended as it reduces the life of the coil.

Note: Do not use solvents to clean the programmable thermostat. Wipe with mild soap and water.

Condenser and Evaporator Fans

DELTA-T air conditioners use high efficiency, long life, sealed ball bearing fans engineered for optimum performance that require no maintenance. Keep fan blades clean for optimal performance. Fans are removable and attached by plug in connections.

Compressor

ISC uses hermetically factory sealed compressors that are quiet, low vibration and are maintenance free. Thermal overload devices are installed to protect the compressor from damage due to overheat and short cycle faults. If the compressor thermal overload fails, it is recommended to return the unit to ISC customer service for proper replacement. Note that the rotary compressor runs with compressed refrigerant gas on the outer case and therefore is hot to the touch.

Loss of Refrigerant

DELTA-T air conditioners are meticulously leak and run tested after assembly. If a leak in the system should develop due to shipping damage or mechanical vibration, the leak must be detected, repaired and the refrigerant charge restored to the system by a qualified refrigeration professional.

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Field Serviceable Parts

ISC carefully designs and selects components with the maximum life expectancy. Due to OEM manufacturing tolerances, poor unit maintenance or extreme operating conditions, components may fail before their maximum life expectancy. The table below lists parts that are serviceable in the field by an Air Conditioner Technician.

Part Description	ISC Part #
Digital Controller	52110-1 for 120VAC 52112-1 for 230VAC/480VAC 52111-1 for 120VAC w/ both Dry C and Heater 52111-2 for 230VAC/480VAC w/ both Dry C and Heater
Low Pressure Switch	38151-1
High Pressure Switch	38176-1
Condenser Fan Cut-In Pressure Switch	38001-1
Temperature Probe, 0.25m	52126-1
Temperature Probe, 1.5m	52126-2-2
Temperature Probe Mounting Clip	52126-250C
Condenser Fan Relay	55761-1 for 120VAC 55762-1 for 230VAC/480VAC
Thermal Expansion Value TXV	RZ-05A/060 = 32002-1, RZ-08A = 32003-1
Compressor	RZ-05A = 31530-1 for 120VAC, 31011-1 for 230VAC/480VAC RZ-06A = 31020-1 for 120VAC, 31011-1 for 230VAC/480VAC RZ-08A = 31020-1 for 120VAC, 31025-1 for 230VAC/480VAC
Compressor Thermal Overload	RZ-05A = 31530-2 for 120VAC, 31011-2 for 230VAC/480VAC RZ-06A = 31020-2 for 120VAC, 31011-2 for 230VAC/480VAC RZ-08A = 31020-2 for 120VAC, 31025-2 for 230VAC/480VAC
Gasket Installation Kit	71403-N568
Evaporator Fan	RZ-05A = 41005-1-1 for 120VAC, 41005-2-1 for 230VAC/480VAC RZ-06A/08A = 41037-1-1 120VAC, 41037-2-1 for 230VAC/480VAC
Condenser Fan	RZ-05A = 41055-1-1 for 120VAC, 41055-2-1 for 230VAC/480VAC RZ-06A/08A = 41045-1-1 120VAC, 41045-2-1 for 230VAC/480VAC
Filter Media	44203 (12pack)
Capacitor	RZ-05A = 54001-1 for 120VAC/230VAC/480VAC RZ-06A/08A = 54004-1 for 120VAC, 54001-1 for 230VAC/480VAC

Safety Information

Unit is carefully designed to restrict access to movable parts to minimize any potential injury. This Air Conditioner has been tested by UL to meet the safety requirements of the UL484 specification. When working with the DELTA-T Air Conditioner always make sure shroud is installed. Verify proper voltage is applied to unit as specified in the UL label of the Air Conditioner. In the event of a field service repair, power down unit using the power on/off switch and disconnect power from unit. It is always recommended to use a licensed Air Conditioner Technician for internal diagnostics and repairs issues.

Contact ISC technical support for further details on opening the unit and troubleshooting tips.

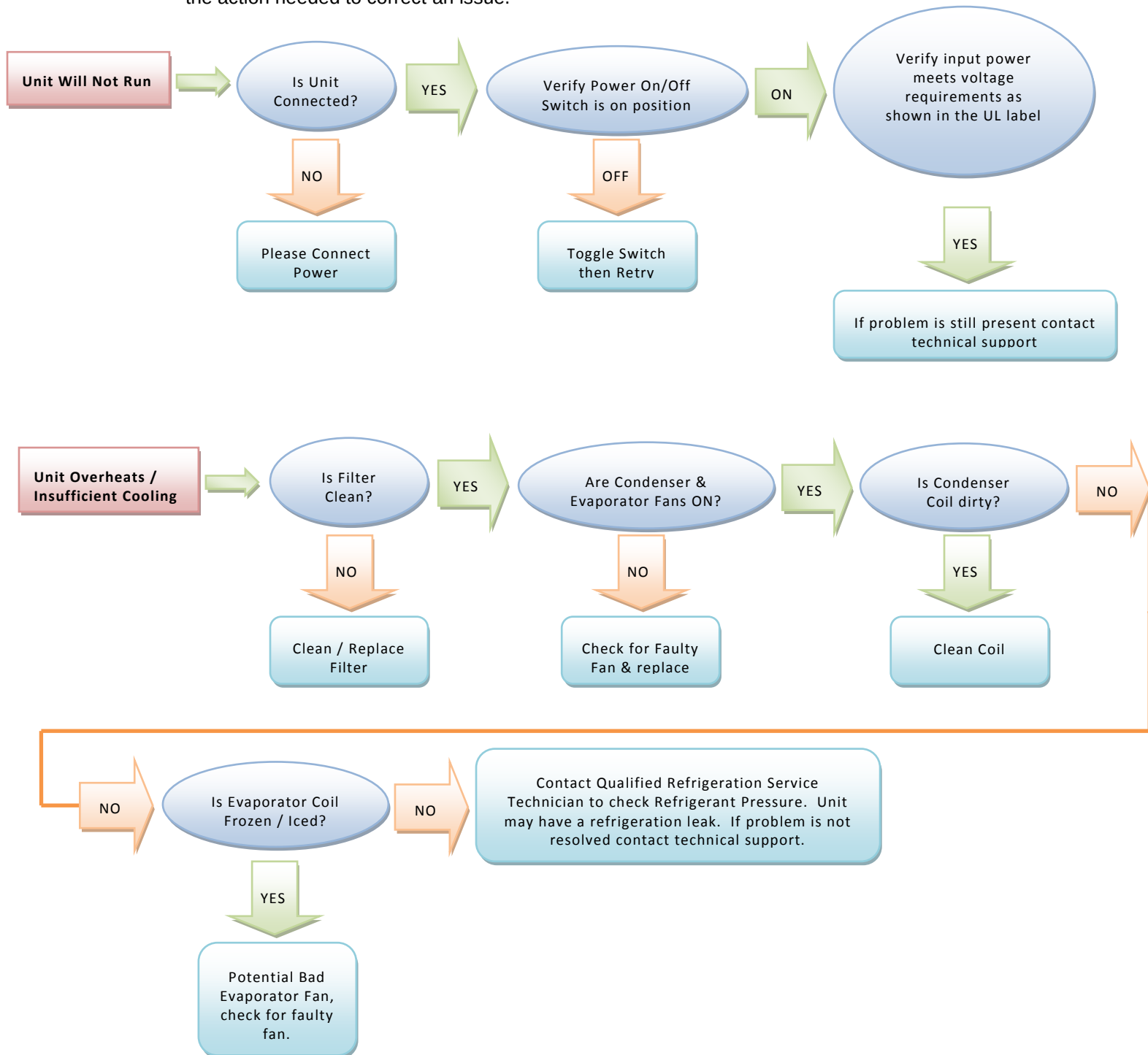
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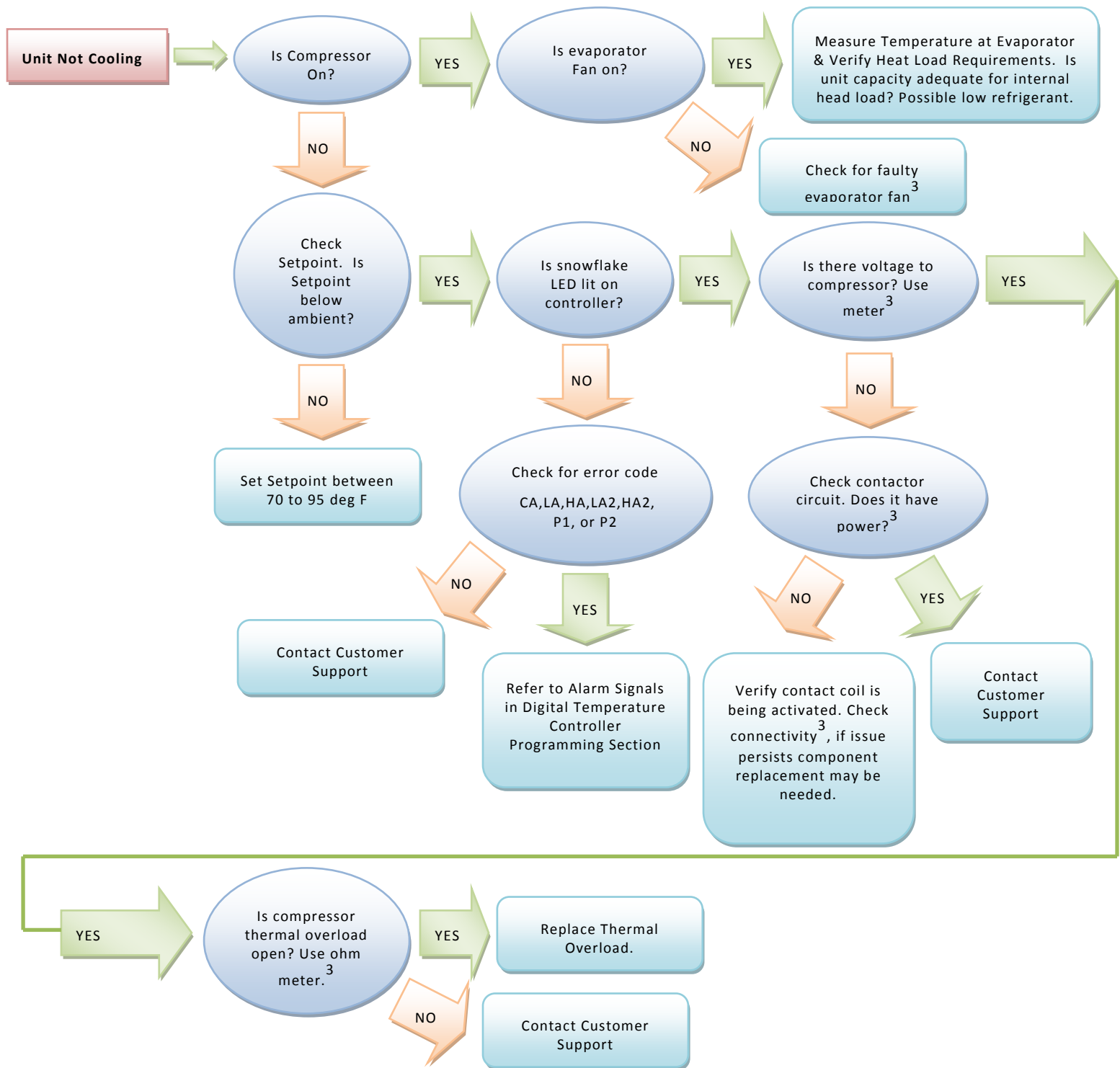
Troubleshooting Guide

The following guide provides a flow chart to identify a problem, determine the root cause and identify the action needed to correct an issue.

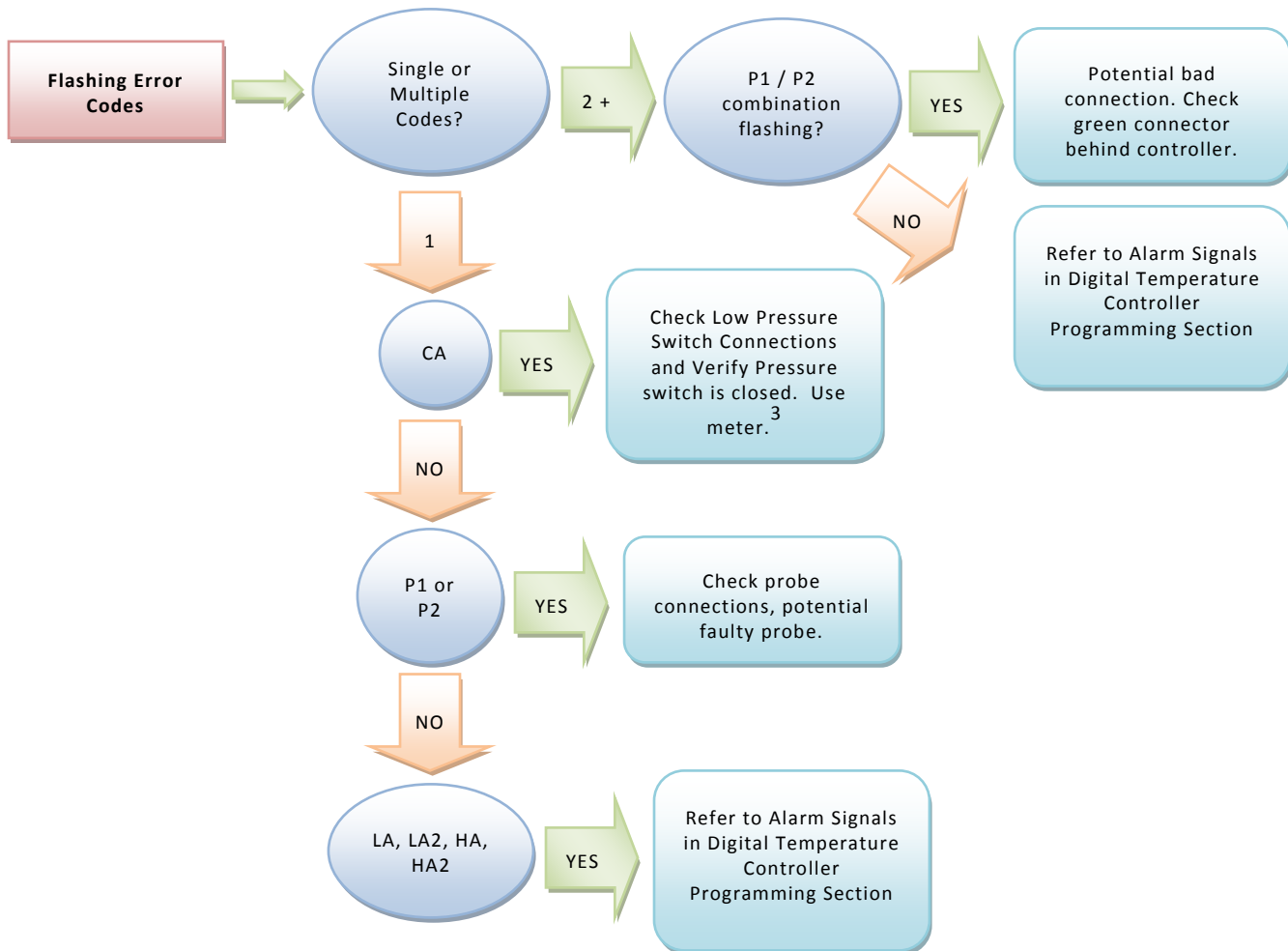


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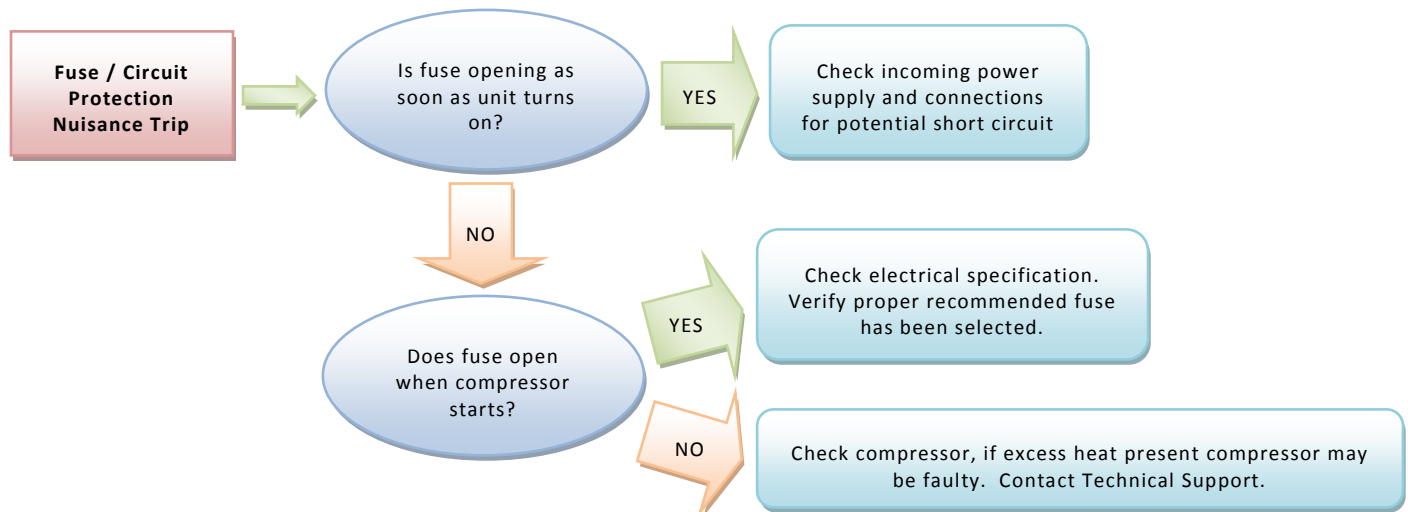
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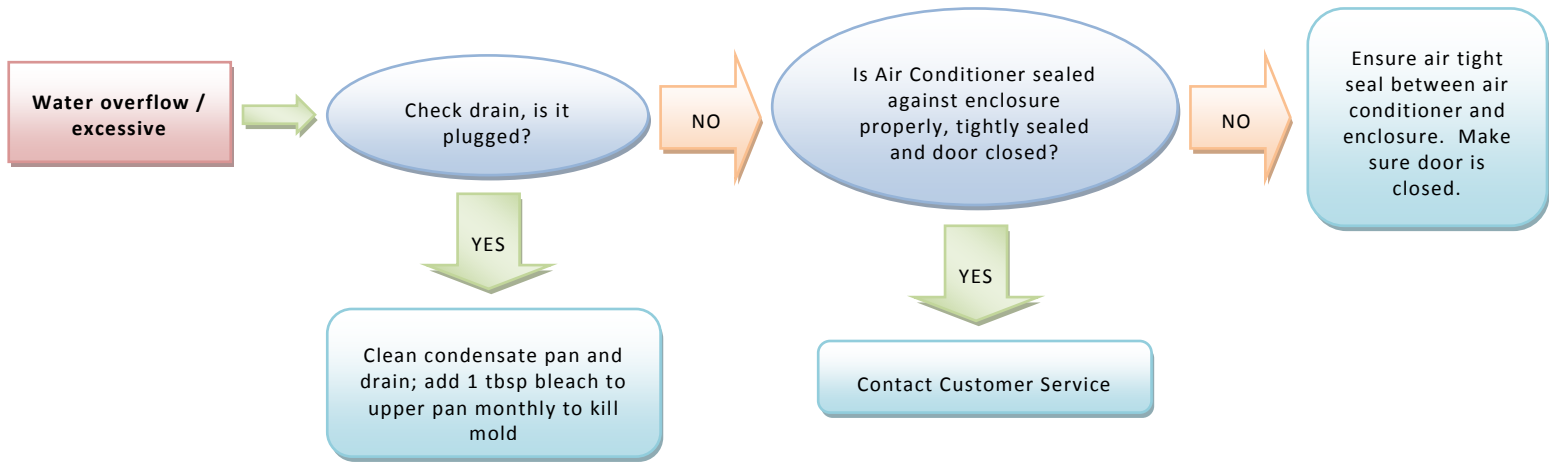


*³Note: Refer to Electrical Schematics for further diagnostics.

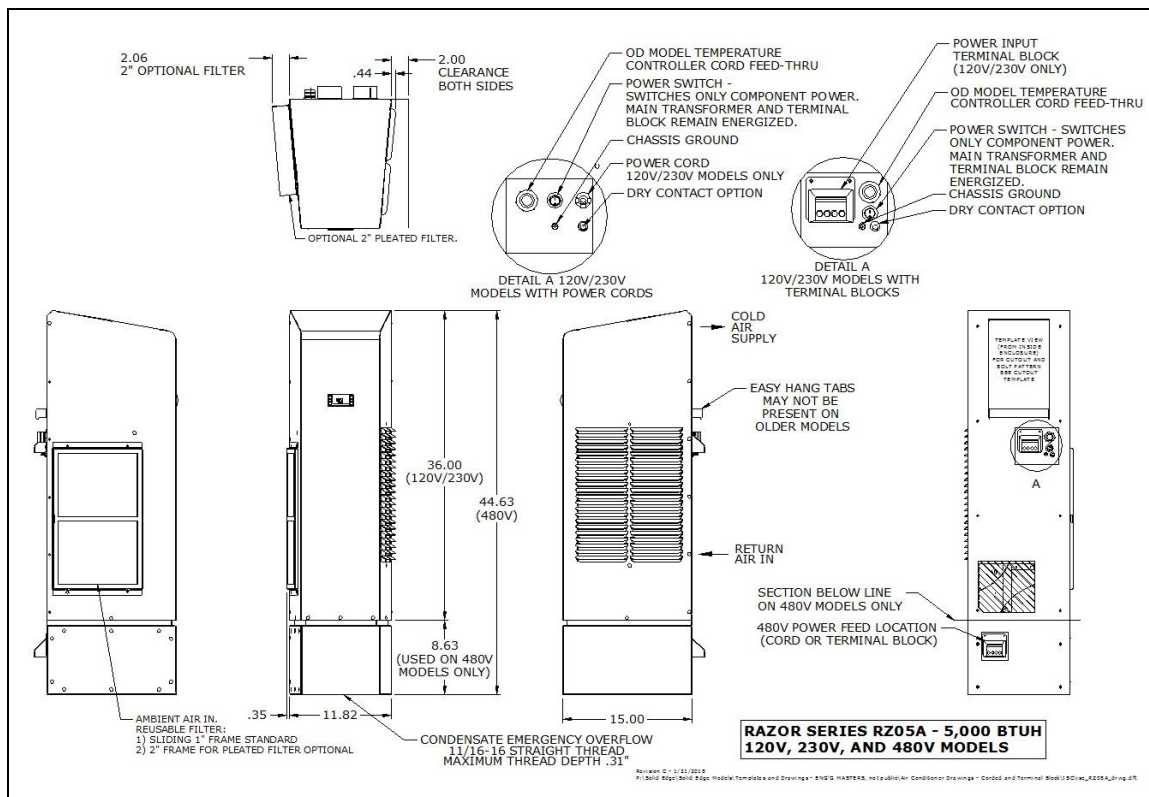


*³**Note:** Refer to Electrical Schematics for further diagnostics.



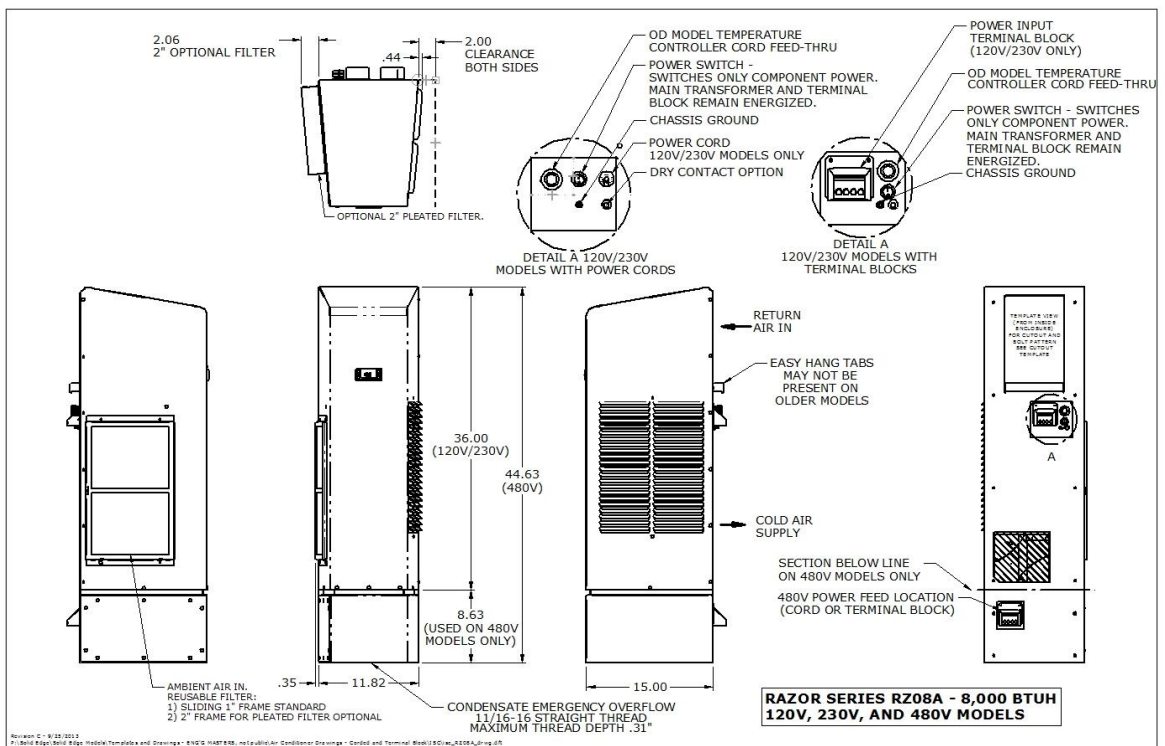
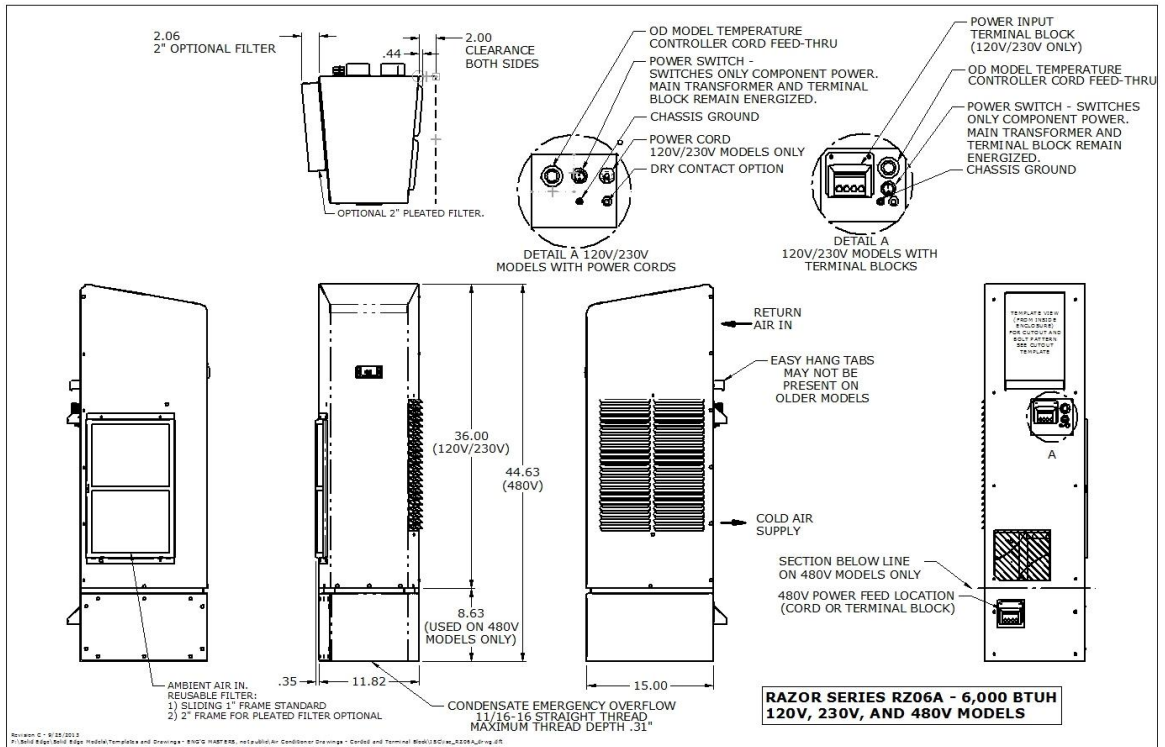


Physical Dimensions



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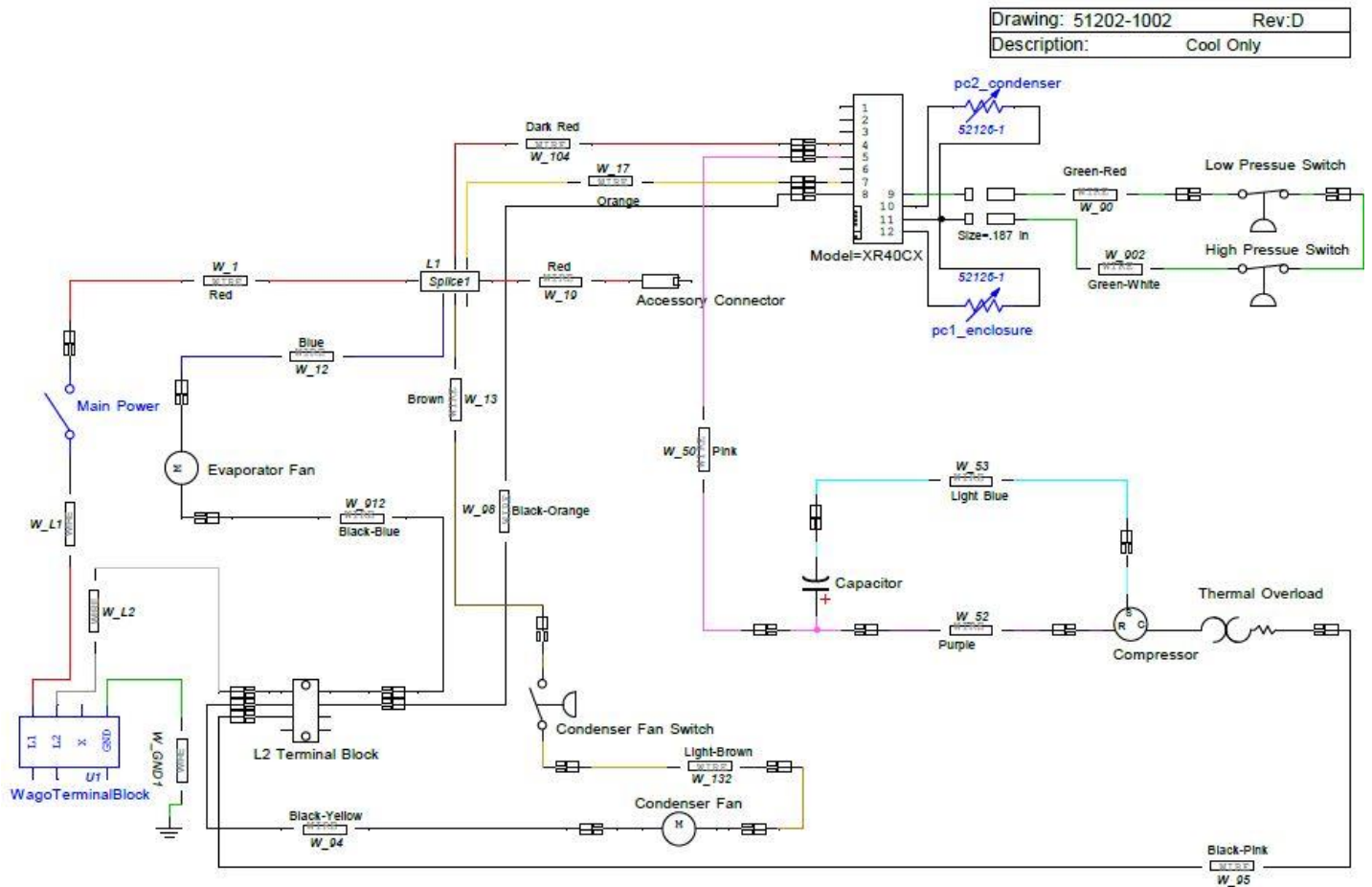


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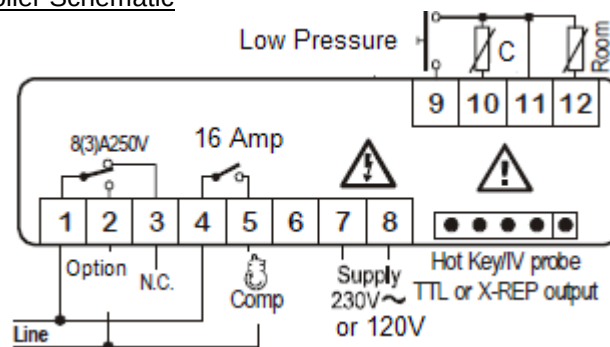
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Electrical Schematics

System Schematic



Digital Controller Schematic



Note: Schematics apply to all voltages; optional accessories wiring not provided in this document. If you require electrical schematic with options contact our technical support and provide your model number.

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Warranty Information

ISC products are warranted to be free of defects in workmanship, materials and components. The warranty period applies from date of shipment for one year. Replacement components have a one year warranty period, except for hermetic system components, which have a 90 day warranty period.

The above warranty applies when the equipment is operated under the following conditions:

- Ambient temperature not in excess of performance rating in normal atmosphere or as stated on product nameplate
- Voltage variation no greater than $\pm 10\%$ from nameplate rating
- Frequency variation no greater than $\pm 3\text{Hz}$ from nameplate rating
- Maximum cooling load no higher than air conditioner nameplate rating
- Waiting five minutes before restarting air conditioner after intentional or accidental shutoff
- Compliance to all other installation, maintenance and operating instructions, as supplied

ISC cannot assume responsibility for misapplication of its products or the erroneous selection of an inappropriate product by a non-authorized ISC representative. Our applications engineers will gladly assist in the selection of the proper product provided all required details of the application are furnished.

ISC assumes no liability beyond the repair or replacement of its own product. This Warranty does not cover:

- Labor or reimbursement of labor for evaluation, removal, installation, repair, or cost of any warranted part, except at the ISC factory in Dallas, Texas
- Use of equipment for other than its designed purpose or operating conditions
- Operation in harsh, oily, corrosive or other abnormal environmental conditions, without the proper filtration, sealing, protective coatings and/or weather protection
- Damage to hermetic system resulting from continuous operation with dirty or clogged air filters or improper or negligent maintenance
- Use of refrigerant other than designated
- Customer modification or abuse
- Shipping damage or other accident
- Repair or service by unauthorized personnel.

Cracked or broken hermetic tubing or brazed joints caused by shipping damage or mishandling are not covered under the Warranty. Claims for shipping damage are the responsibility of the Consignee. Timely claims must be filed with the freight carrier.

The purchaser assumes the responsibility of grounding the unit and installing it in accordance with local electrical and safety codes, as well as the 2008 National Electric Code (NEC) and OSHA.

THIS WARRANTY CONSTITUTES THE ENTIRE WARRANTY WITH RESPECT TO THE PRODUCT AND IS IN LIEU OF ALL OTHERS, EXPRESSED OR IMPLIED, INCLUDING ANY WARRANTY OF MERCHANTABILITY AND WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE AND IN NO EVENT IS ISC RESPONSIBLE FOR ANY CONSEQUENTIAL DAMAGES OF ANY NATURE WHATSOEVER.

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Return Material Authorization (RMA) Procedure

All returns require a Return Material Authorization (RMA) number for warranty or non-warranty repair, rotation of stock, damage or any other reason.



IMPORTANT

Returns without an RMA number will be refused and returned. Improper packaging may void warranty. Air Conditioners shipped laying down will void the warranty. Collect shipments will be refused.

Please be ready to provide:

- Purchase Order Number & Date
- Product Description & Reason for Request
- Model Number & Serial Number
- Customer name and contact info (email, phone number and address)
- Shipping method

Pack unit in a suitable packing for shipment, preferably the original packaging if available.

- If suitable packing is not available, arrange for packaging to be shipped to you.
- Air Conditioners must be returned in an upright position properly secured to a pallet.
- Tip unit to empty water from the evaporator and boil off pans.
- Clearly mark the RMA number on the box.
- Customer will pay all freight charges.

Out of Warranty Repair

If your ISC air conditioner is out of warranty and requires repair, simply call ISC Customer Service at (800)-836-7472 for an RMA number. Customer Service will help you determine what repairs or parts are needed and, if possible, an estimate of the cost.

After the unit is received and diagnosed, you will receive a ship estimate on the work and parts needed. The repairs and test process may uncover other issues for which you will be informed and given quotes for the work needed.



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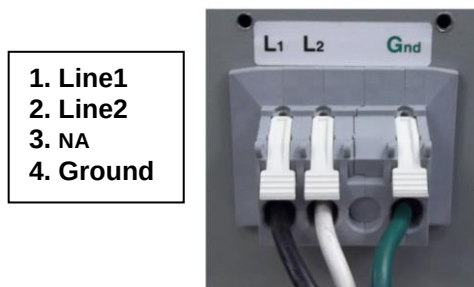


Appendix A

Wago Power Terminal Block Instructions

Scope: Instructions for connecting power wires to air conditioners equipped with a “Wago” power terminal block.

- 1 - Remove cover from chassis using T25 Torx bit.
- 2 -The sequence of the connecting the wires is from LEFT TO RIGHT.
- 3 - Connect each wire as shown by lifting the lever and inserting the wire into the receptacle directly below. Snap the lever down to secure the wire in place.



- 4 – If wires need to be removed, raise the lever directly above each wire to release it.



DESCRIPTION AND INSTALLATION FOR SCANNERS UV1A3, UV1A6, UV8A, UV2, UV2C

DESCRIPTION

The UV1A3/6, UV2/UV2C Scanners are used with the M-Series, M-Series II, MB-2, P, C, and D-Series controls as well as the Fireye FLAME-MONITOR. The UV1A3/6 scanners come with 3 ft. and 6 ft. shielded flexible cables. The UV2 scanner comes with 36 inch (915mm) shielded flexible cable. The UV2 C comes with 36 inch (915mm) unshielded lead wires.

The UV8A Scanner is a 90° right angle ultraviolet flame scanner used in conjunction with the M-Series, M-Series II, MB-2, P, C and D series controls as well as the Fireye® FLAME-MONITOR™. It is available only with 72 inch (1830mm) unshielded lead wires.

The UV tube is made of quartz and is filled with a gas that ionizes when struck by UV radiation from the flame. In the absence of UV radiation the gas acts as an insulator between two electrodes which are mounted inside the tube. These electrodes are energized continuously by a high voltage. During combustion, UV radiation ionize the gas, causing current pulses to flow between the electrodes. These current pulses result in a flame signal which is transmitted to the amplifier in the control, where it is processed to energize or hold in the flame relay.

APPLICATION

UV scanners are recommended for detecting flames from standard fuel gases, some waste gases and light oil fuels such as No. 2 oil. Flames from heavier oils can be detected with UV scanners but the unburned fuel shroud and/or atomization shroud can block passage of the UV from the flame zone to the UV tube, causing nuisance burner/boiler trips. Infrared detection is recommended for fuel oil heavier than No. 2, and some installations where waste gases are burned together with natural gas as a main flame.

NOTE: Certain safety codes and regulations require the use of self-checking ultraviolet scanners. For information on self-checking UV scanners, refer to Bulletin SC-101.

SPECIFICATIONS

Operating Voltage:	(S1-S2): 560-610 VAC 50/60 Hz.
Operating Temperature Limits:	Maximum: 200° F (94° C). Minimum -40° F (-40° C).
Humidity:	85% RH maximum non-condensing.
UV1A3:	36 in. (915mm) shielded leads.
UV1A6:	72 in. (1830mm) shielded leads.
UV8A:	72 in. (1830mm) unshielded leads, 90° right angle mount.
UV2:	36 in. (915mm) shielded leads.
UV2C:	36 in. (915mm) unshielded leads.

IMPORTANT NOTE

The information provided in this bulletin is directed to competent boiler and combustion service technicians who are experienced in the installation and operation of Fireye Flame Safety equipment. Persons not familiar with Fireye products should contact the nearest Fireye representative or other qualified service group.

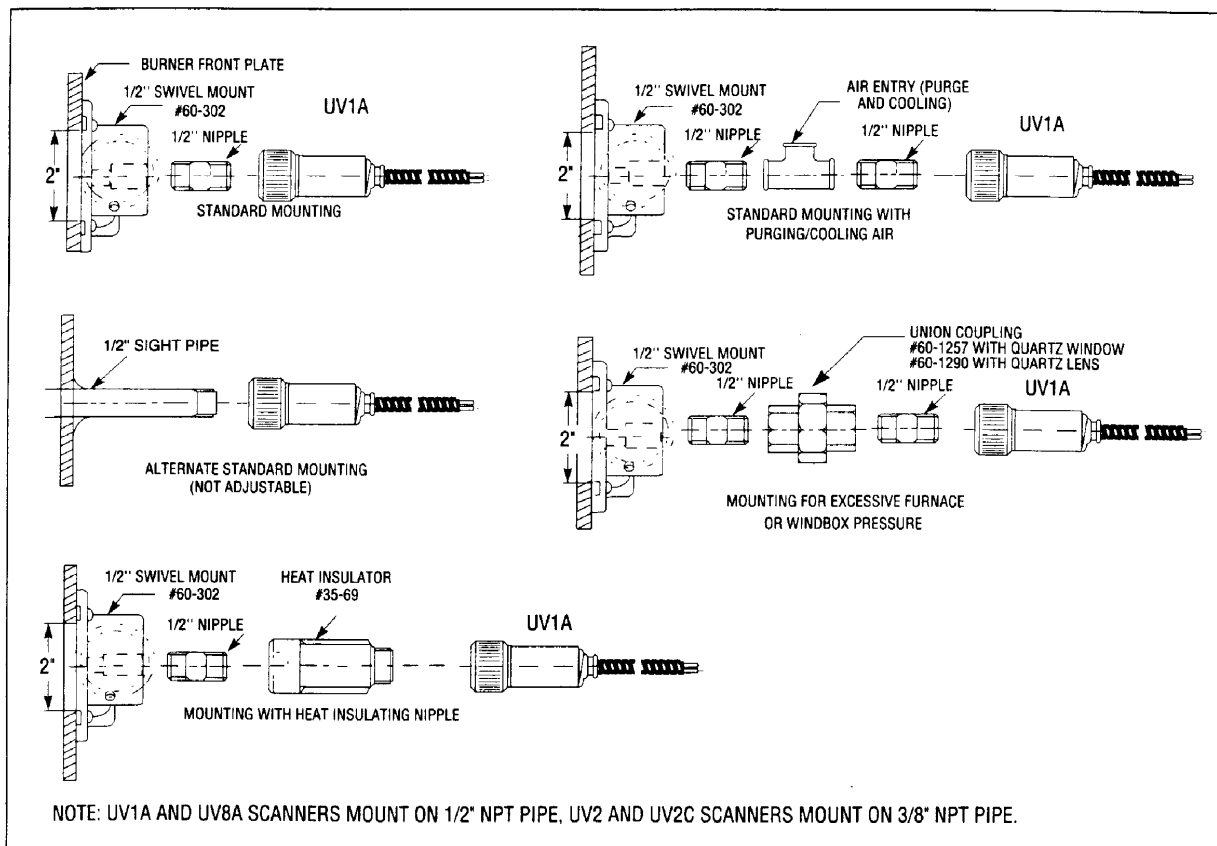
The UV1A, UV2, and UV8A, as well as all other Fireye scanners, are designed to be used exclusively with the appropriate Fireye Flame safeguard and Burner Management controls. The application of these scanners to other than Fireye equipment should be reviewed for approval by Fireye.

APPLICATION REFERENCE

UV8A, UV1A, and UV2, can be used only with the following FIREYE flame safeguard controls.

FIREYE FLAME-MONITOR	C-Series	MB-2 SERIES	M -SERIES M-SERIES II	D SERIES	PFM ADAPTOR KITS
To be used with EUV1 Amplifier Card	25CU6	Use with 61-3359 UV Amplifier	All UVM Series MAU V1 MAUV1T	All D-Series using 72DUV1 Amplifier	PFM2UV PFM3UV

TYPICAL SCANNER MOUNTING





ACCESSORIES

PART NUMBER	DESCRIPTION
60-302	$\frac{1}{2}$ in. Swivel Mount
35-69	Heat Insulator $\frac{1}{2}$ in. NPT threads each end, (male x female).
60-1257	$\frac{1}{2}$ NPT Union with quartz window
60-1290	$\frac{1}{2}$ NPT Union with quartz lens to increase sensitivity
35-75	Water cooled adapter for excessive heat applications

INSTALLATION OF UV1A, UV2, UV8A SCANNERS



WARNING: THE LEADS FROM THE UV8A AND UV2C SCANNER TO THE CONTROL MUST BE SHIELDED TO PREVENT ELECTRICAL NOISE FROM GENERATING A FALSE FLAME SIGNAL TO THE CONTROL.

Where possible, obtain the Burner Manufacturer's Instructions for mounting the scanner. This information is available for most standard burners. The scanner mounting should comply with the following General Instructions:

1. The quartz UV tube is brittle. Care must be exercised to avoid dropping or otherwise striking the scanner with force. Keep the scanner protected until mounted.
2. Position the UV scanner within 18 inches of the flame to be monitored.
3. Select a scanner location that will remain within the ambient temperature limits of the UV Scanner (200°F). If cooling is required, use an insulating coupling (Fireye #35-69) or water cooling adapter (#35-75) to reduce conducted heat.
4. The UV Scanners are designed to seal off against pressures up to 1 PSI. To seal against higher pressures, install a union coupling with quartz window (#60-1257). Add cooling air to reduce the scanner sight pipe temperature.
5. Install the scanner on a standard $\frac{1}{2}$ IPS (3/8 for UV2/UV2C) pipe whose position is rigidly fixed. If the scanner mounting pipe sights through the refractory, do not extend it more than half-way through. Swivel flanges are available if desired (#60-302). The sight pipe must permit an unobstructed view of the pilot and/or main flame, and both pilot and main flames must completely cover the scanner field of view. Oil and gas flames radiate more ultraviolet energy from the base of the flame than from further out in the flame. This fact should be considered when installing the scanner sight pipe.
6. Since the radiated energy from an electric spark ignitor is very rich in the UV region, the field of view should be aimed so that it does not see an electric spark ignitor nor any reflector that is close to the spark.



7. **CAUTION: The unprotected leads must be run in electrical conduit in order to eliminate pickup of electrical noise which could present an unsafe condition.**

8. To prevent spark pickup by the UV detector, close and/or disconnect all pilot and main fuel valves. Operate the control through a burner startup cycle to determine that ignition spark cannot be detected. Burner will not ignite if spark is detected.
9. Smoke or unburned combustion gases absorb ultraviolet radiation. On installations having negative pressure combustion chamber, a small hole drilled in the scanner sight pipe will assist in keeping the pipe clean and free of smoke. For positive pressure furnaces, provide clean air to pressurize the sight pipe, if necessary.
10. Two UV Scanners may be installed on one burner if it is necessary to view two areas to obtain reliable detection of the flame. They should be wired in parallel.
11. To increase scanner sensitivity with the UV Scanners, a quartz lens permits location of the scanner at twice the normal distance. (Union w/quartz lens #60-1290).
12. Request the assistance of any Fireye Field Office for recommendations of a proper scanner installation on a non-standard application.



WIRING-UV SCANNERS

All scanners come with shielded leads except the UV8A and UV2C. The leads for the UV8A and UV2C must be shielded. To connect the scanner to the control, the UV8A Scanner is supplied with 72" lead wire. The UV2C scanner is supplied with 36" of lead wire. If it is necessary to extend the scanner wiring, the following instructions apply:

Scanner wires should be installed in a separate conduit. The wires from several scanners may be installed in a common conduit, but some signal loss can be expected due to capacitive coupling.

Selection of Wire

1. Use #14, 16, or 18 wire with 75° C, 600 volt insulation for up to 100 foot distances (signal loss approximately 20% at 100 feet).
2. For extended scanner wiring up to 500 feet, and for shorter lengths to reduce signal loss, use a shielded wire (Belden 8254, RG-62/U coaxial cable, or equal) for *each* scanner wire. The ends of the shielding must be taped and **not** grounded.
3. Asbestos insulated wire should be avoided.
4. Multiconductor cable is not recommended.



CAUTION: Keep scanner leads well clear of ignition high voltage wiring or any other HV or high current wiring in order to prevent false signal pickup in the scanner leads.

WARNING: Protective filtered lenses should be worn when viewing flame. Infrared and ultraviolet energy can be damaging to the eyes.

MAINTENANCE

Ultraviolet Scanners

The viewing area of the scanner must be kept clean. Even a small amount of contamination on the lens will reduce the flame signal reaching the detector by a measurable amount. Wipe the viewing area routinely using a soft cloth dampened with concentrated detergent, then wipe dry.

TROUBLESHOOTING NON SELF-CHECKING UV SCANNERS

ZERO OR LOW FLAME SIGNAL

1. Zero volts at test jacks on control.
 - Make sure scanner has good view of 1st $\frac{1}{3}$ of flame Reposition scanner for best flame signal. If no signal . . .
2. Check for approximately 560-610 VAC across S1-S2 on control, with no flame present, as follows:
 - Use meter with 20k ohms/volt sensitivity.
 - Deenergize control.
 - Close and secure all fuel valves.
 - Energize control and measure voltage across S1-S2.
3. If voltage across S1-S2 is good . . .
 - With control energized, remove scanner from burner and bring it to within a few inches of a UV source such as a gas lighter or propane torch.
 - Check for flame signal at control.
4. If no flame signal . . .
 - Make sure scanner lens is clean and dry.
 - Deenergize control and check continuity of leads between scanner and control.
 - Replace scanner
 - Replace amplifier or control.

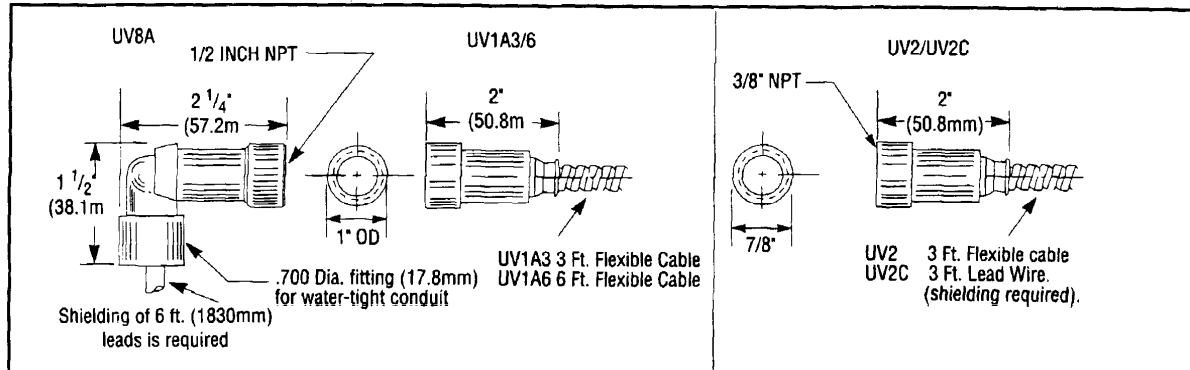


5. If voltage across S1-S2 is low, check L1-L2 for proper line voltage. If line voltage is good:
 - Deenergize control, remove scanner lead wires from terminals S1 and S2.
 - Energize control and recheck for proper voltage across S1 and S2.
 - If voltage across S1 and S2 is good, replace scanner or lead wires.
 - If voltage across S1 and S2 is low, replace amplifier or control.

FALSE FLAME SIGNAL

1. Control will not start burner.
 - Check control wiring, all limits and interlocks.
 - Verify flame signal at test jacks upon energizing control.
2. Flame signal appears at test jacks when no flame is in combustion chamber.
 - Deenergize control to remove scanner leads from S1-S2.
 - Close and secure all fuel valves.
 - Energize control.
 - Check for induced voltage pickup to amplifier.
3. Burner starts up with scanner removed from S1-S2.
 - Deenergize control and connect scanner to S1-S2.
 - Energize control.
 - Check scanner for runaway condition. Look for a blue glow in the UV tube.
 - If glow is present, replace scanner.
 - If no glow is present, check to see that scanner leads are well clear of ignition or other noise carrying lines.
4. Burner does not start up even with scanner removed from S1-S2.
 - Check control wiring, all limits and interlocks.
 - Replace control/amplifier.

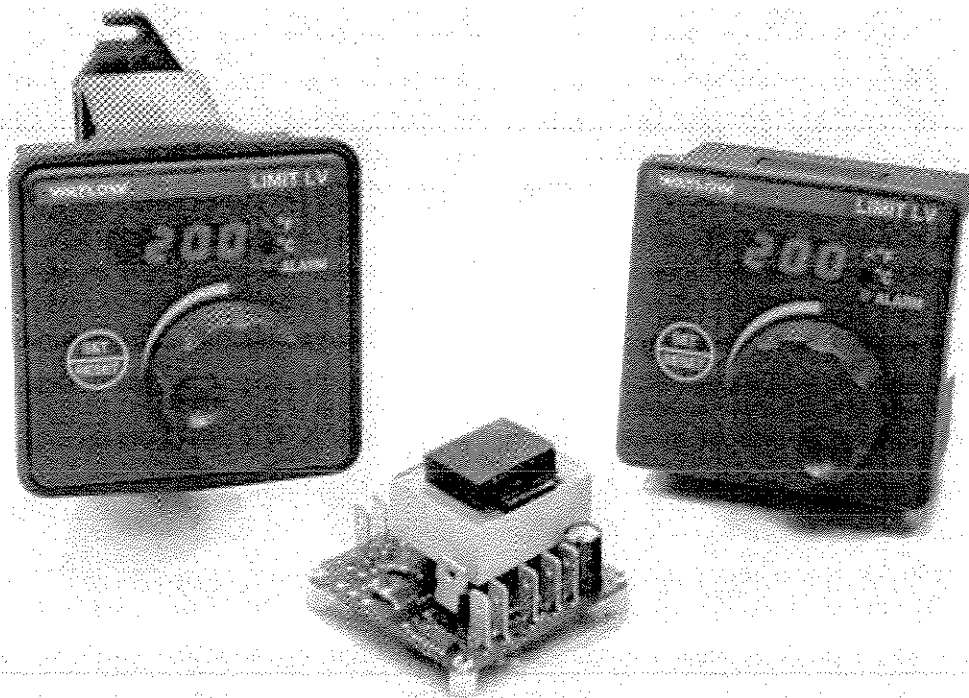
UV SCANNER OUTLINE DIMENSIONS



WARNING: The leads from the UV8A and UV2C Scanner to the control must be shielded to prevent electrical noise from generating a false flame signal to the control.

Series L

User's Manual



Series L - Temperature Limit



1241 Bundy Boulevard., Winona, Minnesota USA 55987
Phone: +1 (507) 454-5300, Fax: +1 (507) 452-4507 <http://www.watlow.com>



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Made in the U.S.A.

\$5.00

Safety Information

We use note, caution and warning symbols throughout this book to draw your attention to important operational and safety information.

A “NOTE” marks a short message to alert you to an important detail.

A “CAUTION” safety alert appears with information that is important for protecting your equipment and performance. Be especially careful to read and follow all cautions that apply to your application.

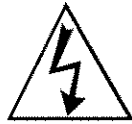
A “WARNING” safety alert appears with information that is important for protecting you, others and equipment from damage. Pay very close attention to all warnings that apply to your application.

The safety alert symbol,  (an exclamation point in a triangle) precedes a general CAUTION or WARNING statement.

The electrical hazard symbol,  (a lightning bolt in a triangle) precedes an electric shock hazard CAUTION or WARNING safety statement.



**CAUTION or
WARNING**



**Electrical
Shock Hazard**

CAUTION or WARNING

Technical Assistance

If you encounter a problem with your Watlow controller, review your configuration information to verify that your selections are consistent with your application: inputs, outputs, alarms, limits, etc. If the problem persists, you can get technical assistance from your local Watlow representative (see back cover), by e-mailing your questions to wintechsupport@watlow.com or by dialing +1 (507) 494-5656 between 7 a.m. and 5 p.m., Central Standard Time (CST). Ask for for an Applications Engineer. Please have the following information available when calling:

- Complete model number
- System wiring information
- Series L Limit User's Manual

Warranty

These controllers are manufactured by ISO 9001-registered processes and are backed by a three-year warranty.

Return Material Authorization (RMA)

1. Call Watlow Customer Service, (507) 454-5300, for a Return Material Authorization (RMA) number before returning any item for repair. We need this information:

- | | |
|--|----------------------------|
| • Ship to address | • Bill to address |
| • Contact name | • Phone number |
| • Method of return shipment | • Your P.O. number |
| • Detailed description of the problem | • Any special instructions |
| • Name and phone number of person returning the product. | |

2. Prior approval and an RMA number, from the Customer Service Department, is needed when returning any unused product for credit. Make sure the RMA number is on the outside of the carton, and on all paperwork returned. Ship on a Freight Prepaid basis.

3. After we receive your return, we will examine it and try to verify the reason for the return.

4. In cases of manufacturing defect, we will enter a repair order, replacement order or issue credit for material returned.

5. To return products that are not defective, goods must be in new condition, in the original boxes and they must be returned within 120 days of receipt. A 20 percent restocking charge is applied for all returned stock controls and accessories.

6. If the unit is unrepairable, it will be returned to you with a letter of explanation.

7. Watlow reserves the right to charge for no trouble found (NTF) returns.

The Series L Limit user's manual is copyrighted by Watlow Winona, Inc., © February 2004 with all rights reserved.

1

Overview

Watlow's Series L family of temperature limit controllers* provide an economical limit controller solution for applications where thermal system protection is needed.

A limit controller is added to applications to prevent over or under temperature conditions. The limit provides safety assurances against instances where a thermal runaway condition occurs as a result of a failed sensor, controller or output device.

The Series L limit controller is recommended for any application where thermal system runaway could result in large product scrap costs, damage to system equipment, potential fire hazard or risk to personnel. All Series L limit controllers are Factory Mutual and CSA approved.

These controllers are available with or without an operator interface and can be ordered in square 1/8th DIN panel mount, din rail mount, open board or potted module design configurations. Push-on, quick connect spade terminal or removable screw clamp style terminal block ordering options provide the electrical connections.

The microprocessor design platform provides improvements in the performance, repeatability, and accuracy offered by Watlow's current line of basic control products.

The Series LV includes an operator interface to allow viewing and selection of the limit set point. A red four character, seven segment LED displays the limit set point. The limit set point selection is made with a continuous turn, velocity sensitive rotary encoder. Push to Set operation reduces accidental limit set point adjustments. Limit set point operating range temperature values are customer definable in the product configuration part number.

The Series LF offers fixed limit set points. These units are supplied without an operator interface. Limit set point temperature values are customer definable in the product configuration part number.

The features and performance of these products make them ideally suited for a wide range of industrial limit control applications in the food preparation, industrial machinery, packaging and plastic markets.

Watlow's Series L limit controllers include industry leading service, support and a 3-year warranty.

*Also available, Series C, an on-off temperature controller version.

Features and Benefits

Four Character LED Display

- Improves limit set point adjustment accuracy.

Fixed or Adjustable Limit Set Points

- Tamper proof operation.
- Control flexibility.

Push to Set

- Reduce accidental limit set point adjustments.

Multiple Mounting Options

- Minimizes installation time.

High or Low Limit Operation

- Application flexibility.

Fahrenheit or Celsius Operation with Indication

- Application flexibility.

Sensor Break Protection

- Provides positive system shutdown.

Agency Approvals

- Meets requirements for agency certification.

Micro Processor Based Technology

- Accurate and repeatable protection.

Stock to Four Day Delivery

2

Installation

Installing the Open Board Controller

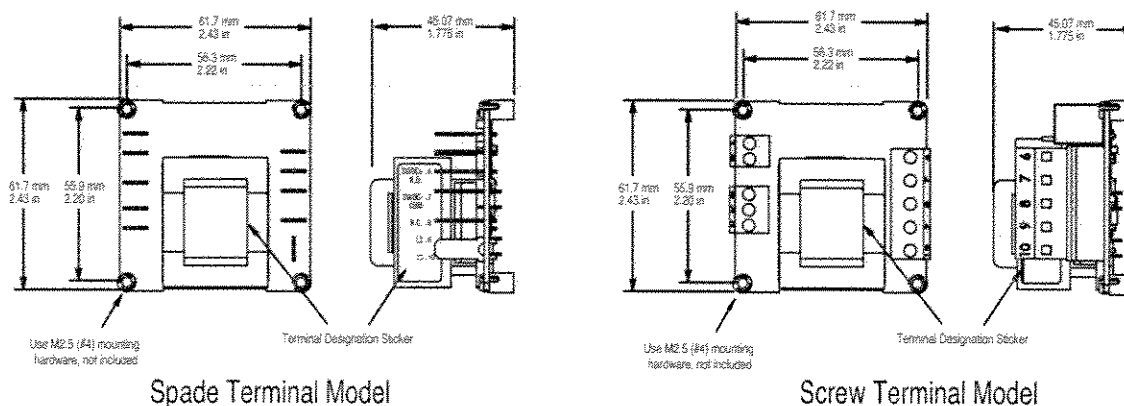


Figure 2a

1. Locate and drill four 3.2 mm (0.125 in) holes in the desired panel location. See Figure 2a for hole locations.
2. Mount the controller using four M2.5 (#4) screws.

Installing the Potted Controller

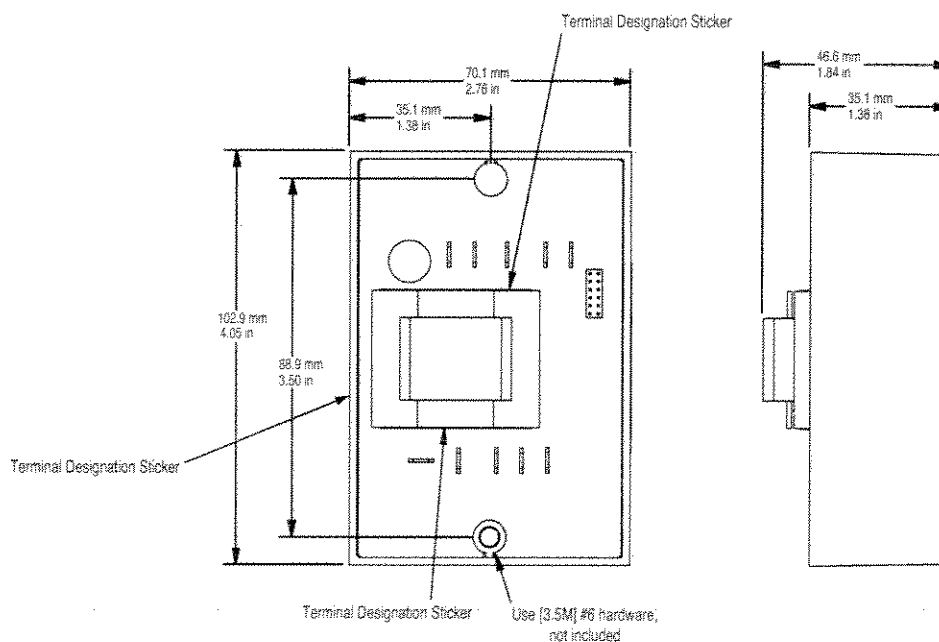


Figure 2b

1. Drill two 5 mm (0.187 in) diameter holes in the desired panel location. See Figure 2b for hole locations.
2. Mount the controller using two M3.5 (#6) screws.

Installing the DIN Rail Mount Controller

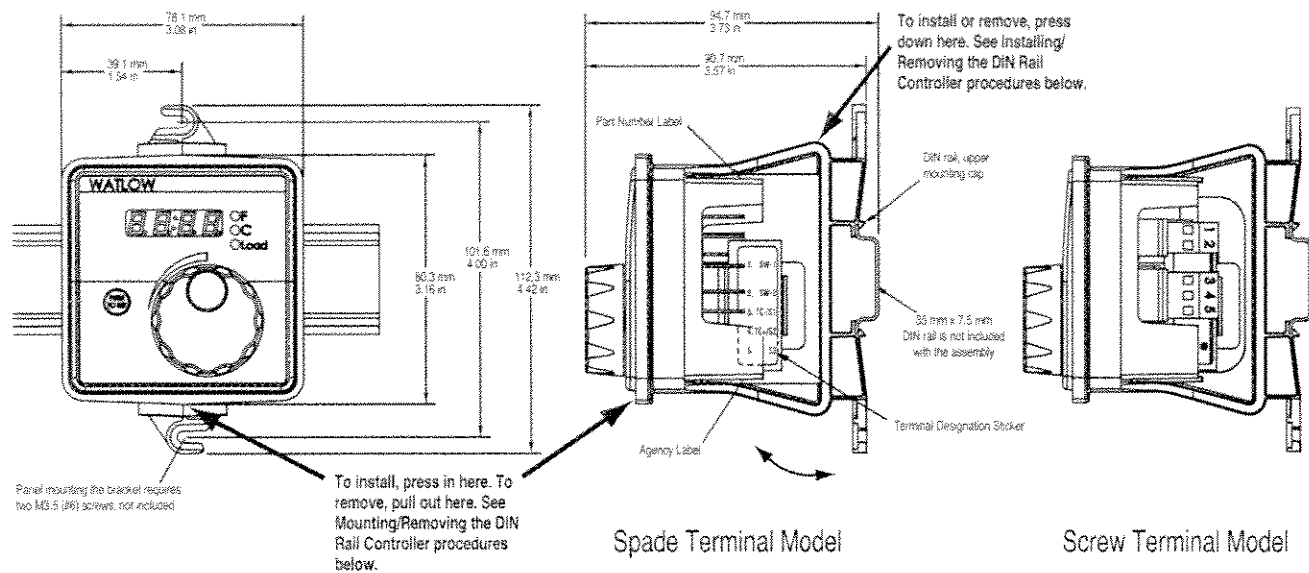


Figure 3a

Sub-Panel Mounting

1. Using the controller as a location template, mark both mounting holes.
2. Drill and tap two 2.7 mm (0.106 in) diameter holes in the desired panel location. See Figure 3a above for hole locations.
3. Mount the controller using two M3.5 (#6) screws.

DIN Rail Mounting

1. Place the DIN rail upper mounting clip on the top edge of the DIN rail. See Figure 3a. DIN rail spec, DIN 50022, 35 mm x 7.5 mm (1.38 in x 0.30 in).
2. Press down firmly on the top back edge of the DIN rail bracket and push in on the bottom, front edge of the bracket. The controller “snaps” securely onto the rail. See Figure 3a. If the controller does not snap on, check to see if the DIN rail is bent. Minimum clipping distance is 34.8 mm (1.37 in), the maximum is 35.3 mm (1.39 in).

Removing the DIN Rail Controller

1. Remove power from the system.
2. Remove all the wiring connections from the back of the controller.
3. While pressing down on the top, back edge of the DIN rail bracket, pull forward on the bottom, front edge of the DIN rail bracket. See Figure 3a.

Removing the Controller from the DIN Rail Bracket

1. Remove power from the system.
2. Remove all the wiring connections from the back of the controller.
3. Remove the DIN rail bracket from the DIN rail.
4. Insert a flat blade screwdriver between the DIN rail bracket and the case. Rotate the screwdriver to release the DIN rail bracket hooks from the ridges on the case, while firmly pushing the controller out the front of the DIN rail bracket. Alternate back and forth between the top and then the bottom. Be sure to support the controller as it comes out of the bracket. See Figure 3b.

Caution: FM approval requires limit switches to be suitably enclosed to restrict casual user adjustment.

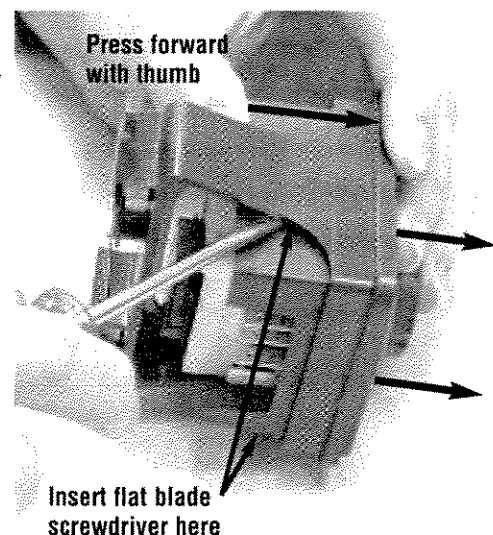


Figure 3b

Installing the Square 1/8 DIN Panel Mount Controller

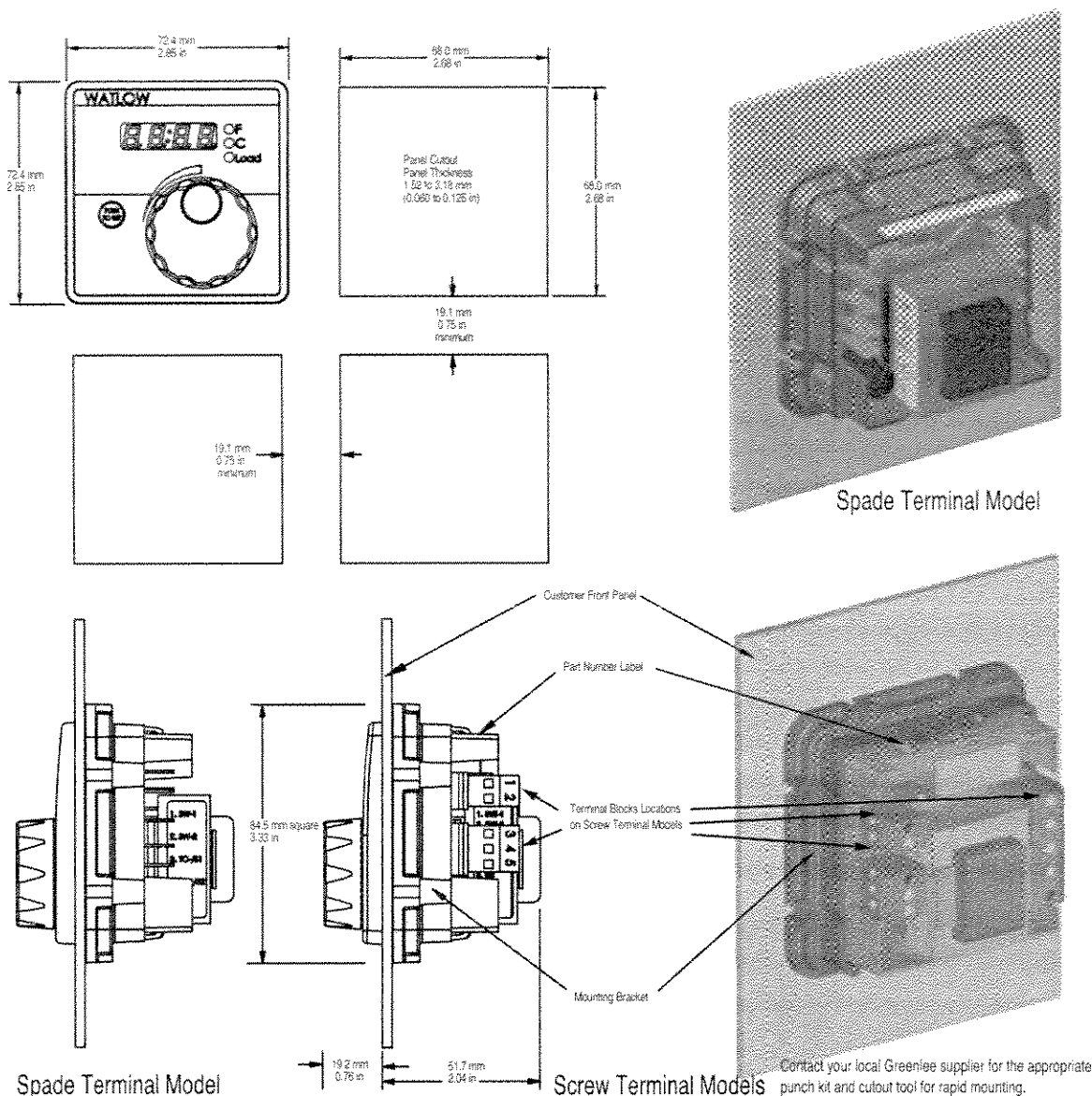


Figure 4

1. Make the panel cutout using the mounting dimensions above.
2. Remove mounting bracket from the back of the controller.
2. Insert the controller into the panel cutout.
3. While pressing the bezel firmly against the panel, slide the mounting bracket over the back of the controller. Be sure the levers on the mounting bracket line up with the teeth on the case.
4. Press the bracket up to the back of the panel. The controller should fit tightly in the panel cutout.

Removing the Panel Mount Square 1/8 DIN Controller

1. Remove power from the system.
2. Remove all the wiring connections from the back of the controller.
3. Slide a thin, wide tool (putty knife) under all three mounting tabs, top then bottom, while pushing forward on the back of the case. Be ready to support the controller as it slides out of the panel cutout.



Caution: FM approval requires limit switches to be suitably enclosed to restrict casual user adjustment.

3

Wiring



Warning:

Use National Electric (NEC) or other country-specific standard wiring and safety practices when wiring and connecting this controller to a power source and to electrical sensors or peripheral devices. Failure to do so may result in damage to equipment and property, and/or injury or loss of life.

Note: Insulated terminals required for quick connect style terminals.

For quick connect terminals 1, 2, 6, 7, 8, 9, and 10, AMP P/N 3-520406-2 or equivalent recommended. Use Amp crimp tool P/N 58078-3, insert 90391-3.

For quick connect terminals 3, 4, and 5, AMP P/N 2-520405-2 or equivalent recommended. Amp crimp tool P/N 58078-3, insert 58079-3.



Caution: FM approval requires limit switches to be suitably enclosed to restrict casual user adjustment.

The terminals on the back of the Series L limits are the same for all of the package styles. They are 6.3 mm (0.25 in) quick connect, push on style terminals or removable screw terminal block. The terminal style is an ordering option.

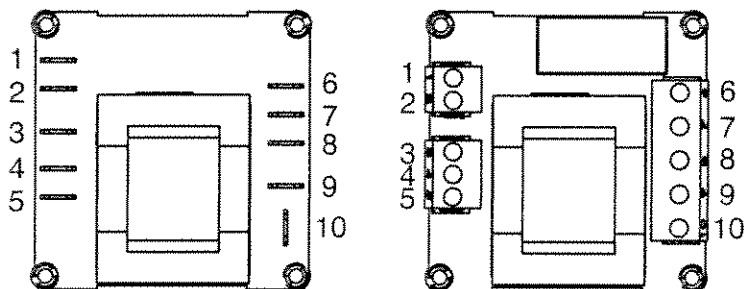
Check the part number to determine your hardware configuration. Refer to the wiring diagrams appropriate for your controller's configuration.

All outputs are referenced to a de-energized state.

Wiring Guidelines

1. Use the correct thermocouple type per the model number on the case sticker of the unit. See dimension drawings for sticker locations.
 - Use correct thermocouple polarity. Red is usually negative.
 - If you must extend thermocouple leads, use thermocouple extension wire to minimize errors.
 - Be sure you have good crimp connections on all wire connections.
 - Insulate the thermocouple mounting from the mounting surface to prevent heat migration input errors.
 - Thermocouple leads should be routed separately from any high voltage lines.
 - Long lead lengths create electrical resistance. When using a two-wire RTD, there will be an additional 2.6° C (4.7° F) error for every 1Ω of lead length resistance. That resistance when added to the resistance of the RTD element, can result in erroneous input to the temperature controller.
2. In electrically-noisy environments (heavy switching contactors, motors, solenoids, etc.), use shielded thermocouple lead wire with the shield connected at the sensor end only.
3. Use a separate thermocouple to maintain the limit function of this controller; do not parallel thermocouple input from the primary controller.
4. All wiring and fusing must conform to the National Electric Code (NEC) NFPA70 and any other locally applicable codes.
5. Fuse the independent load voltage on the L1 (hot) side and connect it to the common (C) side of the relay.

Note: The model number determines the connection terminal style. See below for terminal locations.





Warning:
Use National Electric (NEC) or other country-specific standard wiring and safety practices when wiring and connecting this controller to a power source and to electrical sensors or peripheral devices. Failure to do so may result in damage to equipment and property, and/or injury or loss of life.



WARNING: If high voltage is applied to a low-voltage controller, irreversible damage will occur.

Note: Insulated terminals required for quick connect style terminals.

For quick connect terminals 1, 2, 6, 7, 8, 9, and 10, AMP P/N 3-520406-2 or equivalent recommended. Use Amp crimp tool P/N 58078-3, insert 90391-3.

For quick connect terminals 3, 4, and 5, AMP P/N 2-520405-2 or equivalent recommended. Amp crimp tool P/N 58078-3, insert 58079-3.

Figure 6a — AC Power Wiring

- Nominal voltage options:

- 24V~ (ac)
- 120V~ (ac)
- 230 to 240V~ (ac)

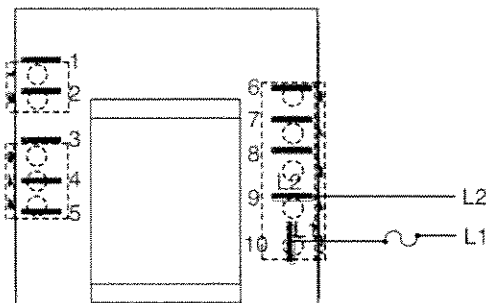


Figure 6b — Thermocouple Input

Thermocouples are polarity sensitive. The negative lead (usually red) must be connected to TC-.

- Input impedance: >10 MΩ

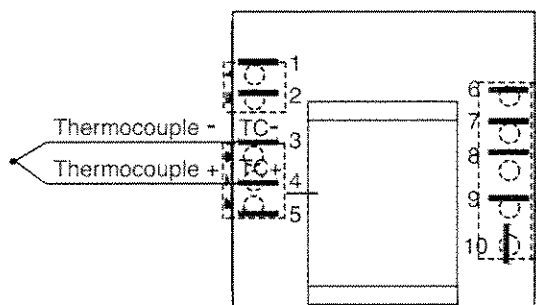
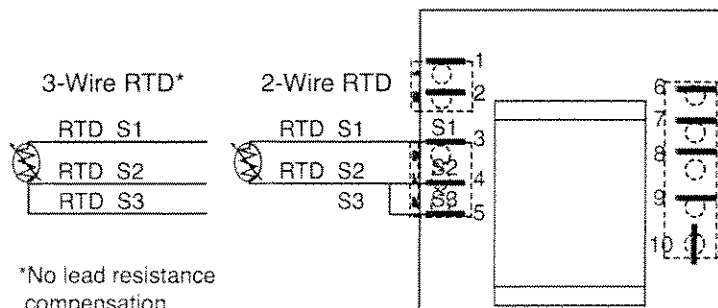


Figure 6c — RTD Input

(100 Ω Platinum DIN curve 0.00385 Ω/Ω/°C)

- Terminals S2 and S3 must be shorted for a two-wire RTD
- Nominal excitation current: 125 μA



*No lead resistance compensation



Warning:

Use National Electric (NEC) or other country-specific standard wiring and safety practices when wiring and connecting this controller to a power source and to electrical sensors or peripheral devices. Failure to do so may result in damage to equipment and property, and/or injury or loss of life.

Note: Use of an external reset switch may affect FM approval. Only the use of a momentary N.O. switch is valid for approval.

Note: Insulated terminals required for quick connect style terminals.

For quick connect terminals 1, 2, 6, 7, 8, 9, and 10, AMP P/N 3-520406-2 or equivalent recommended. Use Amp crimp tool P/N 58078-3, insert 90391-3.

For quick connect terminals 3, 4, and 5, AMP P/N 2-520405-2 or equivalent recommended. Amp crimp tool P/N 58078-3, insert 58079-3.

Quencharc Note:
Switching pilot duty loads (relay coils, solenoids, etc.) with the mechanical relay output option requires use of an R.C. suppressor.

Watlow carries the R.C. suppressor Quencharc brand name, which is a trademark of ITW Paktron. Watlow Part No. 0804-0147-0000.

Figure 7a — **External Reset Switch**

- Momentary normally open (N.O.), dry contact closure

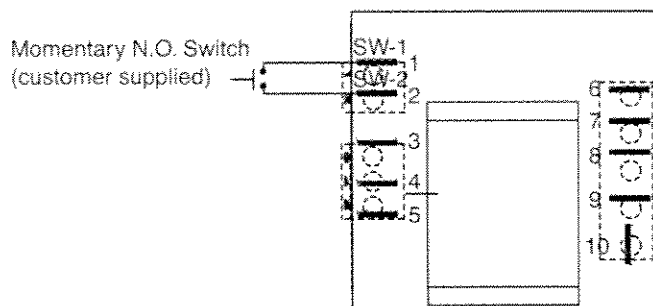
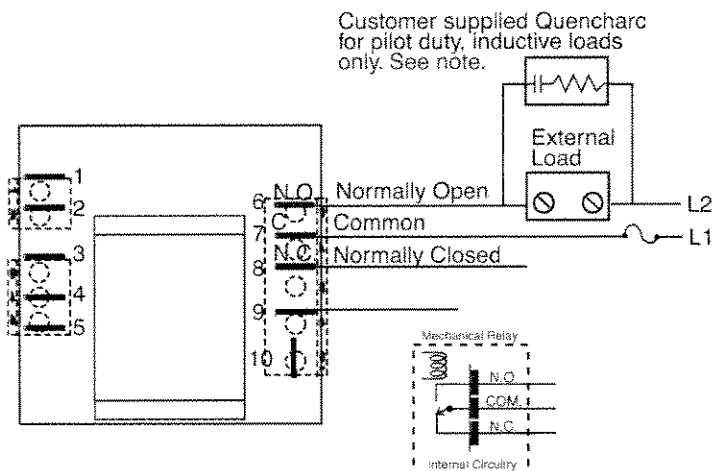


Figure 7b — **Mechanical Relay Output**

- Form C contacts
- 8 A, resistive
- 250 VA pilot duty, 120/240V~ (ac), inductive
- 240V~ (ac) maximum
- 30V= (dc) maximum
- See Quencharc note
- For use with ac or dc
- Minimum load current 100 mA
- Output does not supply power



System Wiring Examples

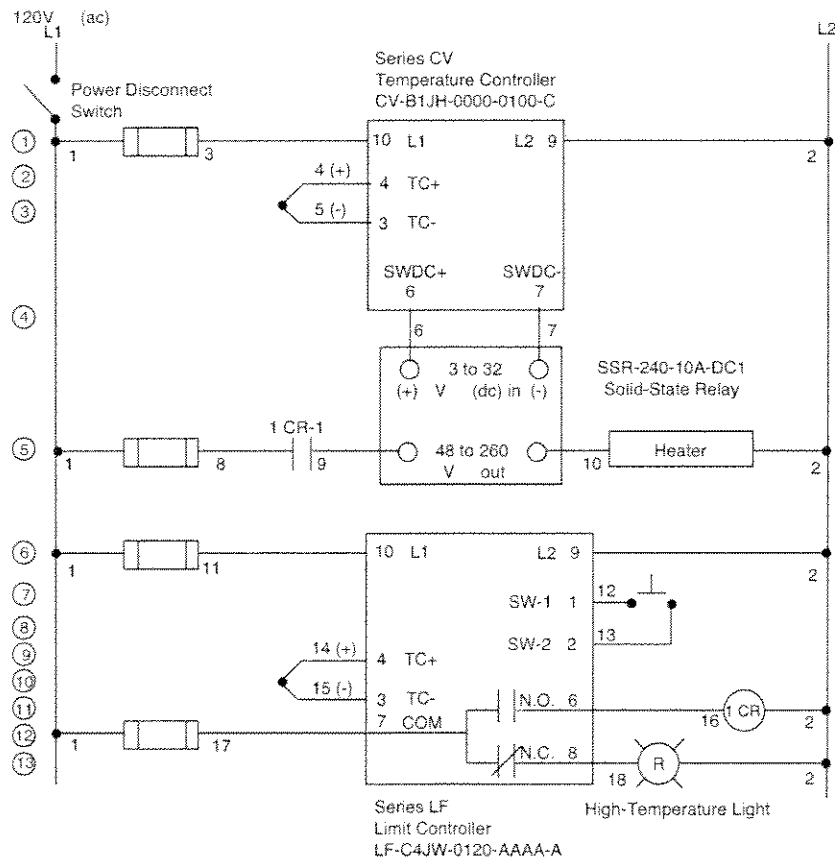
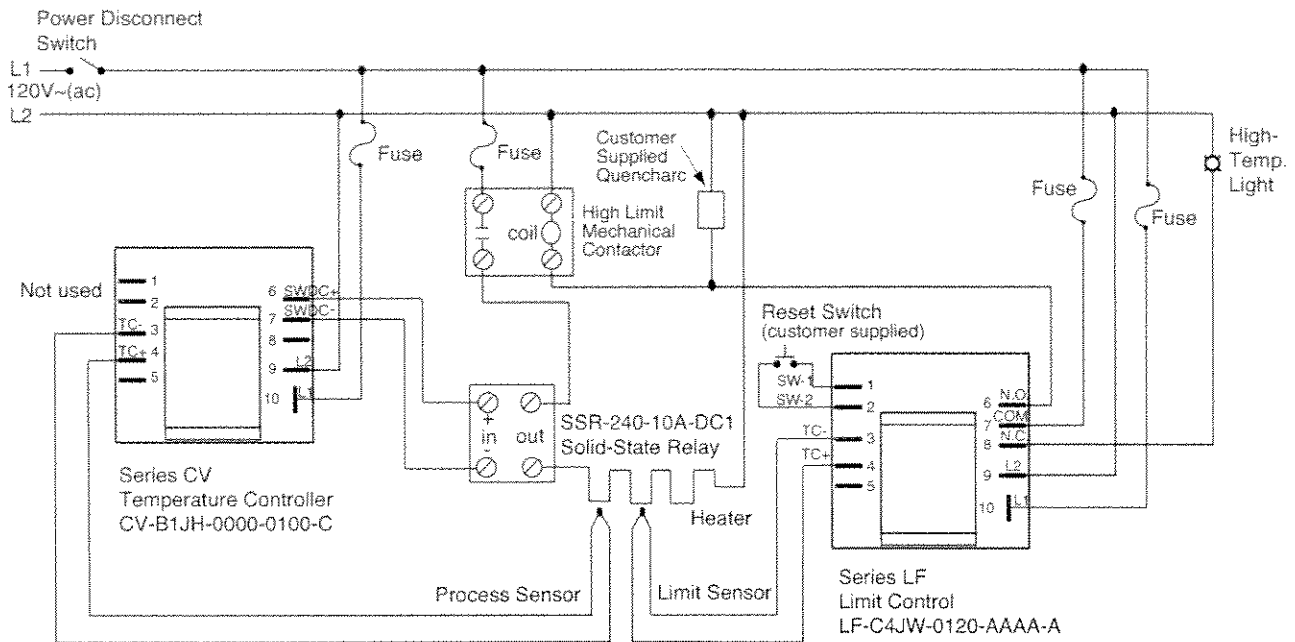


Figure 8 — System Wiring Examples

4

User Interface

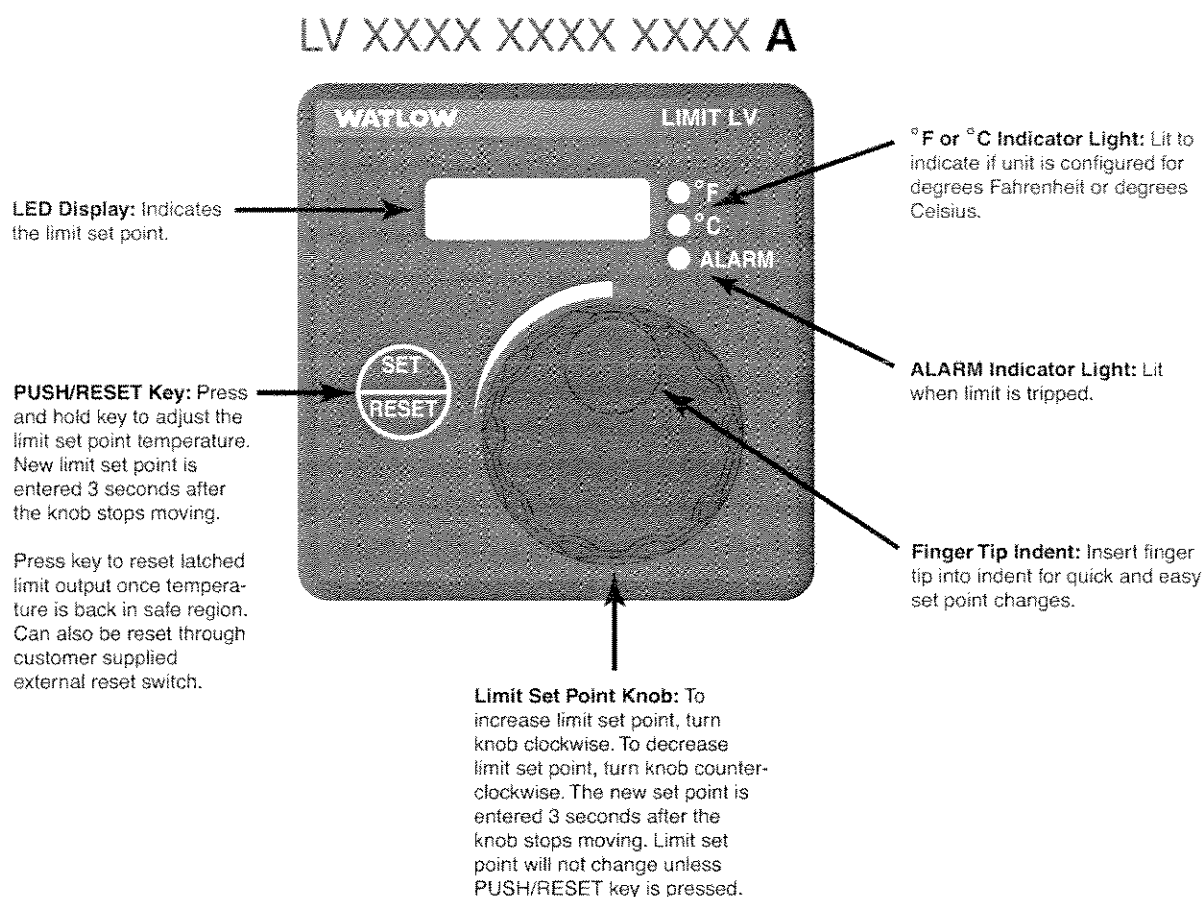


Figure 9 — Variable Limit Set Point, Standard Interface



Caution: FM approval requires limit switches to be suitably enclosed to restrict casual user adjustment.

Troubleshooting

On indicating limits, the display is not illuminated.	• Power supply switch off. • Fuse blown. • Breaker tripped. • Safety interlock door switch activated. • Wiring incorrect or open. • Power supply voltage incorrect. • Defective limit.	• Turn switch on. • Replace fuse (check cause of failure). • Reset breaker (check cause of failure). • Close door. • Check wiring. • Verify input power • Repair or replace limit.
Troubleshooting thermocouple inputs		
Temperature reading is incorrect, showing a sensor error, [Err.Ln] , or ALARM LED is switching at the wrong temperature.	• Sensor or limit may be bad. Sensor connections may be bad. • Ambient temperature in the control cabinet is over 70°C. • Ground loop problem. Can occur when using a switched DC output and a grounded thermocouple.	• Place a jumper wire across the thermocouple input terminals. The display should indicate ambient temperature. If it does, the limit is OK. - For high limit: Start with limit set point above ambient temperature, ALARM LED should be off. Decrease limit set point until ALARM LED goes on. It should be approximately ambient temperature. - For low limit: Start with limit set point below ambient temperature, ALARM LED should be off. Increase limit set point until ALARM LED goes on. It should be approximately ambient temperature. • Measure temperature in cabinet to ensure it is below 70°C. Vent cabinet or add fans if necessary. • Remove power from the system. Use an ohm meter to measure resistance between output DC- and the thermocouple sheath. If there is continuity, replace sensor with an ungrounded thermocouple.
Temperature seems to be decreasing, but actual process is increasing.	• Thermocouple polarity is reversed. In the US, red wire insulation denotes the negative wire.	• Check thermocouple connections. All connections, including extension wire must maintain the correct polarity. Correct polarity problems.
Temperature seems to be reading low and not increasing while actual process temperature is increasing.	• Sensor is bad. Thermocouple is shorted.	• Check thermocouple connections. Check thermocouple wire insulation to make sure it is not damaged, causing the wires to short (making a new junction).
Temperature seems to be off-set from actual process temperature, or the ALARM LED switches on at the wrong temperature. The off-set changes with changes in process temperature.	• Copper wire was used instead of thermocouple extension wire. Connectors of metals different than thermocouple metal were used to splice or make connections.	• Check thermocouple connections. Check to make sure that only thermocouple extension wire of the correct type was used to extend thermocouple leads. Replace if necessary.

Indication	Probable Cause(s)	Corrective Action
	Troubleshooting RTD inputs	
Temperature reading is incorrect, showing a sensor error, ErrLo , or ALARM LED is switching at the wrong temperature.	• Sensor or limit may be bad. Sensor connections may be bad. • Ambient temperature in the control cabinet is over 70°C (158°F). • Sensor connections may be bad. Excessive lead wire resistance.	• Place a 110 ohm resistor across the sensor input terminals. - For high limit: Start with limit set point above ambient temperature, ALARM LED should be off. Decrease limit set point until ALARM LED goes on. It should be approximately ambient temperature. If it does, the limit is OK. Sensor or connections may be bad. - For low limit: Start with limit set point below ambient temperature, ALARM LED should be off. Increase limit set point until ALARM LED goes on. It should be approximately ambient temperature. If it does, the limit is OK. Sensor or connections may be bad. • Measure temperature in cabinet to ensure it is below 70°C (158°F). Vent cabinet or add fans if necessary. • Check sensor connections. Measure lead wire resistance. There will be a 2.6C (4.7°F) error for every ohm of lead wire resistance.
	Troubleshooting limit outputs	
ALARM is not tripped when it should be. ALARM LED is not on (relay is energized in safe condition, N.O. contact is closed and N.C. contact is open).	• Temperature appears to be incorrect. See input troubleshooting. • Limit set point is not set correctly.	• See input troubleshooting. • Verify limit set point setting.
ALARM is tripped when it should not be. ALARM LED is on (relay is de-energized in limit condition, N.O. contact is open and N.C. contact is closed).	• Limit output is tripped (latched). • Output wiring is incorrect. • Temperature appears to be incorrect, see input troubleshooting. • Limit set point is not set correctly. • Limit output is defective.	• Press RESET key to reset limit. • Verify wiring. Relay outputs act as a switch, they do not source power. • See input troubleshooting. • Verify limit set point setting. • Repair or replace limit.
Limit output signal is on when it should not be on. Load LED is on.	• Temperature reading is incorrect on display of indicating controls or limit, see input troubleshooting. • Set point is not set correctly	• See input troubleshooting. • Verify limit set point setting.
Limit load is on when it should be off. ALARM LED is off.	• Power switching device (mechanical relay, contactor, etc.) is shorted. Limit output shorted. • Output wiring is incorrect.	• Remove wires from output of limit to input of power switching device. If load is still on, replace power switching device. If load turns off, replace limit or sensor. See input troubleshooting.. • Verify wiring.

Specifications

Controller

- Microprocessor based, limit controller.
- Nominal switching hysteresis, typically 1.7°C (3°F)
- High or low limit, factory selectable.
- Latching output requires manual reset upon over or under temperature condition.
- Manual or automatic reset on power loss, factory selectable.
- Internal front panel or external customer supplied momentary reset switch.
- Input filter time: 1 second.

Operator Interface (model dependent)

- Four digit, 7 segment LED displays, .28" high.
- °F or °C indicator LED.
- ALARM indicator LED.
- Continuous turn, velocity sensitive rotary encoder for limit set point adjustment.
- Front panel SET/RESET key on variable set point models.
- No operator interface on fixed set point models.

Standard Conditions For Specifications

- Rated line voltage, 50 to 60Hz, 0 to 90% RH non-condensing, 15-minute warm-up.

Calibration ambient range: 25°C (77°F) ±3°C

Sensor Input

Thermocouple

- Grounded or ungrounded.
- Type E, J, K, T thermocouple types.
- >10 MΩ input impedance.
- 250 nV input referenced error per 1Ω source resistance.

RTD

- 2-wire platinum, 100 Ω.
- DIN curve (.00385 curve).
- 125 μA nominal RTD excitation current.

Input Accuracy Span Range

Type E:	-200 to 800°C or -328 to 1,470°F
Type J:	0 to 750°C or 32 to 1,382°F
Type K:	-200 to 1,250°C or -328 to 2,282°F
Type T:	-200 to 350°C or -328 to 662°F
RTD (DIN):	-200 to 800°C or -328 to 1,472°F

Thermocouple Input

- Calibration accuracy: ±1% of input accuracy span, ±1° at standard conditions and actual calibration ambient.
Exception: Type T, ±2.4% of input accuracy span for -200 to 0°C (-328 to 32°F)
- Temperature stability: ±0.3 degree per degree change in ambient.

RTD Input

- Calibration accuracy: ±1% of input accuracy span ±1° at standard conditions and actual calibration ambient.
- Temperature stability: ±0.2 degree per degree change in ambient

Allowable Operating Ranges

External Reset Switch

- Momentary, dry contact closure. See wiring section.

Output Types

Type E:	-200 to 800°C or -328 to 1,470°F
Type J:	-210 to 1,038°C or -346 to 1,900°F
Type K:	-270 to 1,370°C or -454 to 2,500°F
Type T:	-270 to 400°C or -454 to 750°F
RTD (DIN):	-200 to 800°C or -328 to 1,472°F

Electromechanical Relay, Form C

- Minimum load current: 100 mA.
- 8 A @ 240V~ (ac) or 30V= (dc) maximum, resistive.
- 250 VA pilot duty, 120/240V~ (ac) maximum, inductive.
- Use RC suppression for inductive loads.
- Electrical life 100,000 cycles at rated current.

Agency Approvals

Series LE, potted version

- UL®873, UL 197, UL 991 recognized temperature limit for cooking industry. File E43684.
- ANSI Z21.23 Gas appliance thermostat approval.
- CSA C22.2#24 Approved temperature regulator for cooking industry. File #30586.
- FM Class 3545 temperature limit switches. File 3017239.
- CE - see Declaration of Conformity.

Series LE, non-potted version and Series LV

- UL 873 recognized temperature regulator.
- UL 197 reviewed for use in cooking appliances.
- ANSI Z21.23 Gas appliance thermostat approval.
- CSA C22.2#24 approved temperature control. File #30586.
- FM Class 3545 temperature limit switches.
- CE - see Declaration of Conformity.

Terminals

- 6.4 mm (0.25 in) quick connect, push-on terminals. See order options. Refer to Wiring section for crimp-on terminal recommendations.
- Removable screw clamp style terminal blocks. See order options.
- Wire gauge 0.1 to 4 mm² (30 to 12 AWG). Strip length, 8 mm (0.30 in).
- Torque: 0.8 Nm (7 in-lb) maximum.

Power

- 24V~ (ac) +10%; -15%; 50/60 Hz, $\pm 5\%$
- 120V~ (ac) +10%; -15%; 50/60 Hz, $\pm 5\%$
- 230 to 240V~ (ac) +10%; -15%; 50/60 Hz, $\pm 5\%$
- 10VA maximum power consumption.
- Data retention upon power failure via nonvolatile memory.

Operating Environment

- 0 to 70°C (32 to 158°F)
- 0 to 90% RH, non-condensing.
- Storage temperature: -40 to 85°C (-40 to 185°F)

Dimensions

- DIN Rail model can be DIN rail or chassis mount
DIN rail spec, DIN 50022, 35 mm x 7.5 mm (1.38 in x 0.30 in)

Style	Width	Height	Depth
Open board	61.7 mm (2.43 in)	61.7 mm (2.43 in)	45.1 mm (1.78 in)
Potted	70.1 mm (2.76 in)	102.9 mm (4.05 in)	46.6 mm (1.84 in)
DIN Rail	78.1 mm (3.08 in)	112.3 mm (4.42 in)	90.7 mm* (3.57 in)
Square 1/8 DIN Panel	72.4 mm (2.85 in)	72.4 mm (2.85 in)	Behind panel 51.7 mm (2.04 in)

*Depth including DIN rail, 94.7 mm (3.73 in)

Note: These specifications are subject to change without prior notice.

Glossary

automatic power reset — A feature in latching limit controllers that does not recognize power outage as a limit condition. When power is restored, the output is re-energized automatically, as long as the temperature is within limits.

latched output — Limit control output latches in de-energized condition when over or under temperature condition occurs and cannot be reset unless temperature drops below set point.

limit or limit controller — A highly reliable, discrete safety device (redundant to the primary controller) that monitors and limits the temperature of the process or a point in the process. When temperature exceeds or falls below the limit set point, the limit controller interrupts power through the load circuit. A limit controller can protect equipment and people when it is correctly installed with its own power supply, power lines, switch and sensor.

manual power reset — A feature in latching limit controllers that does recognize power outage as a limit condition. When power is restored, the output is not re-energized automatically, even if the process is within limits. An operator must use the reset key or switch to manually re-energize the output on power up.

manual reset — A feature on a limit controller that requires human intervention to return the limit to normal operation after a limit condition has occurred.

safety limit — An automatic limit intended for use in applications where an over-temperature fault may cause a fire.

Ordering Information and Model Numbers

Limit Control no user interface

L F

A A A A

Set Point Type

F Fixed Limit Set Point

Line Voltage

C 120V~ (ac)

E 230-240V~ (ac)

G 24V~ (ac)

Controller Package

1 Panel Mount, Square 1/8 DIN -Spade Terminals

2 DIN Rail Mount -Spade Terminals

3 Open Board, not potted -Spade Terminals

4 Potted Case -Spade Terminals

5 Panel Mount, Square 1/8 DIN -Screw Terminals

6 DIN Rail Mount -Screw Terminals

7 Open Board, not potted -Screw Terminals

Sensor and Scale

H Type J -346 to 1900 Degrees F

J Type J -210 to 1038 Degrees C

K Type K -454 to 2500 Degrees F

L Type K -270 to 1370 Degrees C

M Type T -454 to 750 Degrees F

N Type T -270 to 400 Degrees C

P 100Ω RTD -328 to 1472 Degrees F

R 100Ω RTD -200 to 800 Degrees C

S Type E -328 to 1470 Degrees F

T Type E -200 to 800 Degrees C

Limit Type

U High Limit, Manual Power Reset

W High Limit, Auto Power Reset

Y Low Limit, Manual Power Reset

Z Low Limit, Auto Power Reset

Fixed Limit Set Point Value *

XXXX Limit Set Point Value**

Overlay/Custom Option

A Standard

*Note: Limit set point must fall within the sensor operating range.

**Note: A (-) is used in the left digit of the operating range to indicate negative values.

Ordering Information and Model Numbers

Limit Control, LED display, Front panel reset switch **L V**

Set Point Type

V Variable Limit Set Point

Line Voltage

C 120V~ (ac)

E 230-240V~ (ac)

G 24V~ (ac)

Controller Package

1 Panel Mount, Square 1/8 DIN -Spade Terminals

2 DIN Rail Mount -Spade Terminals

5 Panel Mount, Square 1/8 DIN -Screw Terminals

6 DIN Rail Mount -Screw Terminals

Sensor and Scale

H Type J -346 to 1900 Degrees F

J Type J -210 to 1038 Degrees C

K Type K -454 to 2500 Degrees F

L Type K -270 to 1370 Degrees C

M Type T -454 to 750 Degrees F

N Type T -270 to 400 Degrees C

P 100Ω RTD -328 to 1472 Degrees F

R 100Ω RTD -200 to 800 Degrees C

S Type E -328 to 1470 Degrees F

T Type E -200 to 800 Degrees C

Limit Type

U High Limit, Manual Power Reset

W High Limit, Auto Power Reset

Y Low Limit, Manual Power Reset

Z Low Limit, Auto Power Reset

Low Limit Set Point Range Limit *

XXXX Low Limit Set Point Operating Range Value**

High Limit Set Point Range Limit *

XXXX High Limit Set Point Operating Range Value

Overlay/Custom Option

A Standard

*Note: Set point ranges must fall within the sensor operating range.

**Note: A (-) is used in the left digit of the operating range to indicate negative values.

Notes

Declaration of Conformity

Series L

**Watlow Winona, Inc.
1241 Bundy Blvd.
Winona, MN 55987 USA**



Declares that the following product:

Designation: **Series L**
Model Numbers: **LF** - (C, E or G) (1, 2, 3, 4, 5, 6 or 7) (any letter) (U, W, Y or Z) - (any four numbers or - and three numbers) - (AAAA) - may be followed by additional numbers or letters
LV - (C, E or G) (1, 2, 5 or 6) (any letter) (U, W, Y or Z) - (any four numbers or - and three numbers) - (any four numbers) - may be followed by additional numbers or letters
Classification: LF and LV = Temperature Regulator Installation Category II, Pollution degree 2
Rated Voltage: 24, 120, 230/240 V~ (ac)
Rated Frequency: 50 or 60 Hz
Rated Power Consumption: 10VA maximum

Meets the essential requirements of the following European Union Directives by using the relevant standards shown below to indicate compliance.

89/336/EEC Electromagnetic Compatibility Directive

EN 61326: 1997 With A1, 1998: **Electrical equipment for measurement, control and laboratory use – EMC requirements (Industrial Immunity, Class B Emissions).**
A2:2001:
EN 61000-4-2: 1996 + A1, 1998 Electrostatic Discharge Immunity
EN 61000-4-3: 1997 Radiated Field Immunity
EN 61000-4-4: 1995 Electrical Fast-Transient / Burst Immunity
EN 61000-4-5: 1995 + A1, 1996 Surge Immunity
EN 61000-4-6: 1996 Conducted Immunity
EN 61000-4-11: 1994 Voltage Dips, Short Interruptions and Voltage Variations Immunity
EN 61000-3-2: 1995 + A1-3: 1999 Harmonic Current Emissions
EN 61000-3-3: 1995 + A1, 1998 Voltage Fluctuations and Flicker

73/23/EEC Low-Voltage Directive

EN 60730-1:2000 With A11:2002 and **Automatic electric controls for household and similar use:**
EN 60730-2-9:2002: **Particular requirements for temperature sensing controls.**
LXX4 units
EN 61010-1:2001 **Safety Requirements of electrical equipment for measurement,**
All other units **control and laboratory use. Part 1: General requirements**

Raymond D. Feller III
Name of Authorized Representative

Winona, Minnesota, USA
Place of Issue

General Manager
Title of Authorized Representative

February 2004
Date of Issue

Signature of Authorized Representative

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